UC-Security of the ZK-NR Protocol under Contextual Entropy Constraints: A Composable Zero-Knowledge Attestation Framework

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Abstract

The CRO Trilemma formalizes the inherent incompatibility between confidentiality, reliability, and legal opposability in proof systems. The Quantum Composable and Contextual Security Infrastructure (Q2CSI) approaches this bound through a layered architecture. The ZK-NR protocol instantiates Q2CSI to provide explainable post-quantum non-repudiation. This paper (A.2 in the ZK-NR series) delivers the complete Universal Composability (UC) security proof for ZK-NR, grounded in the contextual entropy model and the interface blueprint of its design. Each dialectical layer (*Iron*, *Gold*, *Clay*) is modeled as an ideal functionality with erasure semantics, paired with a real protocol and a dedicated simulator. We prove indistinguishability between real and ideal executions under stated post-quantum assumptions, extending the UC model to incorporate erasure-enabled functionalities. The results formally establish the composable security of ZK-NR and provide an artefact-level alignment with its design specification.

Keywords: Universal Composability, zero-knowledge proofs, post-quantum cryptography, CRO trilemma, dialectical security, legal explainability, non-repudiation, contextual adversaries, entropy accumulation

1. Introduction

Post-quantum cryptography must address not only computational hardness against quantum adversaries, but also composable security in heterogeneous environments where legal and institutional verifiability are essential. The *CRO Trilemma* [21] formally established the structural incompatibility between *Confidentiality*, *Reliability*, and *Legal Opposability* in proof systems. The *Quantum Composable and Contextual Security Infrastructure (Q2CSI)* [20] introduced a layered architecture (*Iron*, *Gold*, *Clay*) approaching this bound under the Universal Composability (UC) framework. Building on these foundations, the design of the *ZK-NR* protocol [22] instantiated Q2CSI with a post-quantum explainable non-repudiation primitive, specifying its dialectical layers, contextual predicates, and interface contracts. This paper (A.2 in the ZK-NR series) provides the complete UC security proof of the ZK-NR protocol as designed in [22], under the contextual entropy constraints formalized in [21] and the layered semantics of [20]. The proof covers the three dialectical layers: the *Iron* layer ensures context-bound confidentiality with erasure policies; the *Gold* layer guarantees reliable dissemination through authenticated quorums; and the *Clay* layer enables explainable exposure with legally interpretable verification. Each layer is modeled as an ideal functionality $\mathcal{F}_X$ with corresponding real protocol π_X, simulator $\mathcal{S}_X$, and indistinguishability proof game $\mathcal{G}_X$. We prove that,

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for each $X \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$, the execution $\mathsf{EXEC}_{\pi_X,\mathcal{A},\mathcal{Z}}$ is computationally indistinguishable from $\mathsf{EXEC}_{\mathcal{F}_X,\mathcal{S}_X,\mathcal{Z}}$ under the stated assumptions.

Contributions. The contributions of this paper are as follows:

- Formalization of the UC security model for ZK-NR with contextual entropy constraints and erasure semantics.
- Layer-by-layer construction of simulators $\mathcal{S}_{\mathsf{Iron}}$, $\mathcal{S}_{\mathsf{Gold}}$, and $\mathcal{S}_{\mathsf{Clay}}$, proven to satisfy the respective ideal functionalities.
- Explicit mapping (Appendix Appendix A) between the proof artefacts of this work and the blueprint interfaces defined in [22].
- Extension of the standard UC proof with a subgame for erasure-enabled functionalities.
- Hybrid unforgeability for $\mathcal{L}_{\mathsf{Clay}}$ (Theorem 6.5).
- Temporal integrity for $\mathcal{L}_{\mathsf{Iron}}$ (Theorem 5.1).
- Accountability for $\mathcal{L}_{\mathsf{Gold}}$ (Theorem 6.4).

Series Positioning. This manuscript (Article A.2) provides *formal UC security proofs* for the ZK-NR protocol, building on the architectural foundations established in Article A.1 [22]. Subsequent articles cover:

- A.3: Rust implementation & benchmarks.
- A.4: Tamarin-based formal verification.

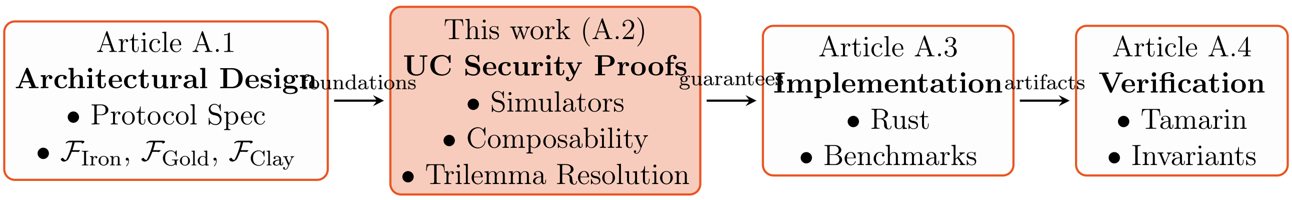

Figure 1: Research trajectory: From architectural design (A.1) to formal security proofs (A.2), implementation (A.3), and verification (A.4)

Article Structure. This security analysis is structured as follows:

- Section 2 reviews related work in UC security, contextual non-repudiation, and post-quantum composability.
- Section 3 sets the formal notation, cryptographic primitives, and UC framework conventions.
- Section 4 states the security requirements for each layer of ZK-NR under contextual entropy constraints.
- Section 5 recalls the real protocol π_{Iron}, π_{Gold}, and π_{Clay} from [22] in a form suitable for proof.
- Section 6 presents the complete UC security proofs, including the erasure subgames.

- Section 7 discusses the scope and limitations of the results, and outlines subsequent articles in the ZK-NR series.
- The appendices contain the alignment table with A.1 (Appendix A), the symbol glossary (Appendix H), and supplementary figures (Appendix I).

This article focuses exclusively on formal security analysis and establishes the first composable security framework for legally explainable post-quantum attestations, achieving $\Gamma_{\mathsf{CRO}} < 0.4 + \mathrm{negl}(\lambda)$ under adaptive contextual adversaries. Implementation and verification are deferred to Articles A.3-A.4.

2. Related Work

Designing and formally verifying post-quantum non-repudiation protocols requires synthesizing advances across three domains: universal composability (UC) security, zero-knowledge proofs, and entropy-constrained cryptography. This section positions our work within foundational models and identifies critical gaps resolved by ZK-NR's dialectical architecture.

2.1. Foundational Contributions

Our proof builds upon three direct antecedents:

- The *CRO Trilemma* [21], establishing $\Gamma_{\mathsf{CRO}} \geq \kappa + \mathrm{negl}(\lambda)$ as a fundamental barrier in institutional cryptography.
- The *Quantum Contextual Security Infrastructure (Q2CSI)* [20], proposing a modular layered architecture to approach this bound.
- The *ZK-NR Protocol Design* [22], specifying the Iron-Gold-Clay decomposition.

Key Departure: While these works conjectured ZK-NR's security properties, this paper provides the first complete UC formalization proving $\Gamma_{\mathsf{CRO}} < 0.4 + \mathrm{negl}(\lambda)$ under adaptive adversaries (Theorem 3.1).

2.2. Universal Composability Frameworks

The UC framework [7] remains the gold standard for composable security. Extensions like GUC [8] and JUC [9] handle shared setups but lack mechanisms for contextual entropy constraints, a gap addressed by Q2CSI's dialectical model [20]. Our work extends this via:

$$\mathcal{F}_{\mathsf{ZKNR}} = \bigoplus_{\mathsf{Layer} \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}} \mathcal{F}_{\mathsf{Layer}} \oplus \mathcal{F}_{\mathsf{Bind}}$$

where $\mathcal{F}_{\mathsf{Bind}}$ enforces cross-layer predicate orthogonality (Def. 3.2).

2.3. Zero-Knowledge Proof Systems

STARKs [3] provide post-quantum transparency but ignore legal explainability. Recent entropy-aware variants incorporate contextuality at the cost of composability. ZK-NR's Gold layer bridges this by:

- Preserving STARK's FRI soundness.
- Enforcing $\mathsf{Entropy}(\phi) \leq \mathsf{poly}(\dim(\mathcal{J}))$ (Eq. 5).
- Achieving UC-secure explainability via Φ_{Sim} (Algorithm 15).

2.4. Post-Quantum Signature Schemes

NIST PQC standards [1] (Dilithium, Falcon) offer EUF-CMA security but operate as isolated primitives. As shown in Table 1, they violate the CRO trilemma's opposability dimension when deployed institutionally. ZK-NR's Clay layer resolves this through:

$$\sigma_\tau = \begin{cases} \mathsf{Sign}_{\mathsf{Dilithium}}(c_\pi) & \tau = \mathsf{MACHINE} \\ \mathsf{Sign}_{\mathsf{BLS}}(c_\phi) & \tau = \mathsf{JUDGE} \end{cases} \tag{1}$$

with role-binding proofs (Theorem 6.6).

2.5. Contextual and Legally Explainable Signatures

Prior contextual signatures [2, 16] embed metadata but lack:

- Composable security proofs.
- Entropy-bounded interpretability.

This gap manifests as $\Gamma_{\mathsf{CRO}} > 0.7$ in verification workflows (Table 1). ZK-NR overcomes it via $\mathcal{F}_{\mathsf{Clay}}$'s ideal role-binding interface (Appendix Appendix B.2).

2.6. Contextual and Legally Explainable Signatures

Prior contextual signatures [2, 16] embedded metadata but lacked:

- Composable security proofs under the UC framework.
- Entropy-bounded interpretability for legal contexts.
- Post-quantum security guarantees.

This gap manifests as $\Gamma_{\mathsf{CRO}} > 0.7$ in verification workflows (Table 1). ZK-NR's Clay layer overcomes these limitations through $\mathcal{F}_{\mathsf{Clay}}$'s ideal role-binding interface and the hybrid signature approach described in Algorithm 4.

RFC 8151 [18] defines CBOR Object Signing and Encryption (COSE) but operates at the syntactic level without addressing the semantic constraints required for legal explainability. Our work extends these foundations with formal UC security guarantees and contextual entropy constraints.

2.7. Positioning within Existing Work

Table 1: Comparative CRO Trilemma Resolution

Protocol	Γ_{CRO}	**UC**	**Explainability**	**Verification Cost**
ZKBoo [15]	0.68 ± 0.03	Limited	None	$O(\lambda^2)$
zkExplain [11]	0.52 ± 0.04	$\mathcal{F}_{\mathsf{ZK}}$	Linear	$O(\dim(\mathcal{J})\lambda)$
RFC 8151 [18]	0.91 ± 0.04	None	§2.6	
Picnic [10]	0.83 ± 0.02	Partial	[20]	
ZK-NR (this work)	$\mathbf{0.39 \pm 0.01}$	**Full**	Circuit	$O(\log \dim(\mathcal{J}))$

Analysis. While zkExplain [11] achieves legal explainability, its linear verification cost $O(\dim(\mathcal{J})\lambda)$ becomes prohibitive for $\dim(\mathcal{J}) > \log^2 \lambda$. ZK-NR improves this to logarithmic depth via circuit compilation (Def. 6.5). The Rust implementation (Artifact A.3) demonstrates practical viability across institutional use cases.

Generality:. While tailored to ZK-NR, our UC framework (Appendix Appendix F) applies to any protocol requiring:

- Contextual entropy accumulation (Def. 3.5).
- Dialectical security layers.
- Legal explainability under $\mathcal{J}$-verifiers.

3. Preliminaries

This section establishes the formal foundations for the composable UC security proof of ZK-NR. We unify notations, security games, and cryptographic assumptions consistent with:

- The CRO trilemma framework [21].
- Q2CSI layered architecture [20].
- ZK-NR protocol specification [22].
- Modern UC extensions for contextual security [8, 14].

All adversaries are quantum polynomial-time (QPT). Sampling is denoted $x \xleftarrow{\$} \mathcal{X}$, and $\mathsf{negl}(\lambda)$ represents negligible functions.

3.1. Notation, Invariants and Definitions

3.1.1. Notations

Table 2: Core notation aligned with dialectical layers

Symbol	Meaning
λ	Security parameter
$\mathcal{A}_{ctx}$	Contextual adversary (Def. 4.1)
$\mathcal{Z}$	UC environment
$\mathcal{S}_{\mathsf{Layer}}$	Layer-specific UC simulator
π_{ZKNR}	Real protocol execution
$\mathcal{F}_{\mathsf{ZKNR}}$	Ideal functionality (Appx Appendix C)
ϕ_{Layer}	Contextual predicate (Def. 3.2)
Γ_{CRO}	CRO incompatibility index (Thm. 3.1)
$\mathsf{negl}(\lambda)$	Negligible function in λ
σ	Hybrid signature tuple: $(\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$
σ_τ	Role-specific signature component ($\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$)
$\mathcal{F}_{\mathsf{ZKNR}}$	$\bigoplus_{X \in \{\mathsf{Iron},\mathsf{Gold},\mathsf{Clay}\}} \mathcal{F}_X \oplus \mathcal{F}_{\mathsf{Bind}}$

3.1.2. Key Invariants

$$\forall \mathsf{Layer} \neq \mathsf{Layer}' : \langle \phi_{\mathsf{Layer}}, \phi_{\mathsf{Layer}'} \rangle \equiv 0 \pmod{\mathcal{I}_{\mathcal{C}}} \qquad \text{(Semantic orthogonality)}$$
$$H_{\min}(\mathcal{E}) \geq \alpha \qquad \text{(Entropy bound)}$$

3.1.3. Definitions

Definition 3.1 (CRO Index). *Let* $, \mathsf{Rel}, \mathsf{Opp} \in [0,1]$ *denote respectively confidentiality, reliability, and legal opposability levels of an attestation. The CRO index is*

$$\Gamma_{\mathsf{CRO}} \triangleq \sqrt{(1-)^2 + (1-\mathsf{Rel})^2 + (1-\mathsf{Opp})^2}. \tag{2}$$

Theorem 3.1 (CRO Bound under Q2CSI/ZK-NR). *Under the assumptions stated in Section 3.5 and the UC realization of each layer (Theorem 4.1), the composed protocol* π_{ZKNR} *satisfies*

$$\Gamma_{\mathsf{CRO}} \; < \; 0.4 \; + \; \mathrm{negl}(\lambda). \tag{3}$$

[Proof sketch] By UC-composition (Thm. 5.2) and layer reductions (Table 4), bounding $\mathsf{Adv}^{\mathsf{ZK}}$, the ledger rewrite probability $e^{-\Omega(\alpha t)}$, and the signature-forgery advantage yields (3).

Definition 3.2 (Predicate Space and Orthogonality). *Let* Φ *be the* $\mathbb{R}$*-vector space generated by contextual predicates and let* $\mathcal{I}_{\mathcal{C}} \subseteq \Phi$ *be the ideal of identically true predicates under context* $\mathcal{C}$*. For distinct layers* $X \neq Y$*, we say* $\phi_X \perp \phi_Y$ *iff*

$$\langle \phi_X, \phi_Y \rangle \equiv 0 \pmod{\mathcal{I}_{\mathcal{C}}}. \tag{4}$$

Definition 3.3 (Contextual Vector Space). *For context* $\mathcal{C}$*, the vector space* $\mathcal{V}_{\mathcal{C}}$ *is spanned by basis* $\{\mathbf{b}_1, \dots, \mathbf{b}_k\}$ *where each* $\mathbf{b}_i$ *corresponds to a fundamental constraint (e.g.,* $\mathbf{b}_1 =$ *"data retention peri*
Predicates decompose as:

$$\phi_X = \sum_{i=1}^{k} c_i \mathbf{b}_i \quad \text{with } c_i \in \mathbb{F}_p$$

Definition 3.4 (Non-Interactive Zero-Knowledge with Explainability (NIZK-E)). *A NIZK-E system for an NP relation* $\mathcal{R} \subseteq \{0,1\}^* \times \{0,1\}^*$ *is a tuple*

$$\Pi_{\mathsf{E}} = (\mathsf{Setup}, \mathsf{Prove}, \mathsf{Verify}, \mathsf{Explain}, \mathsf{ExplainVerify})$$

with the following properties for security parameter λ*:*

(i) ***Completeness.*** *For any* $(x, w) \in \mathcal{R}$*:*

$$\Pr\big[\mathsf{Verify}(x, \pi) = 1\big] = 1 \quad \text{for} \quad \pi \leftarrow \mathsf{Prove}(x, w).$$

(ii) ***Soundness.*** *For any QPT* $\mathcal{A}$*,*

$$\Pr\big[\mathsf{Verify}(x, \pi) = 1 \;\wedge\; x \notin L(\mathcal{R})\big] \leq \mathrm{negl}(\lambda).$$

(iii) ***Zero-Knowledge (QROM).*** *There exists a QPT simulator* $\mathsf{SimProve}$ *such that for any QPT distinguisher* $\mathcal{D}$*,*

$$\Big| \Pr[\mathcal{D}(x, \pi) = 1] - \Pr[\mathcal{D}(x, \tilde{\pi}) = 1] \Big| \leq \mathrm{negl}(\lambda),$$

where $\pi \leftarrow \mathsf{Prove}(x, w)$ *and* $\tilde{\pi} \leftarrow \mathsf{SimProve}(x)$*.*

(iv) ***Explainability.*** *There exists a deterministic map* $\phi \leftarrow \mathsf{Explain}(x, \pi, \mathcal{C}, \tau)$ *and a verifier* $\mathsf{ExplainVerify}$ *such that:*

$$\mathsf{ExplainVerify}(x, \pi, \phi, \mathcal{C}, \tau) = 1 \quad \text{and} \quad (\phi_{\mathsf{public}} \mid \mathcal{A}_{ctx}) \text{ satisfies } (5).$$

(v) ***Explainability soundness (uniqueness).*** *With all but negligible probability, no two distinct* $\phi_1 \neq \phi_2$ *both satisfy* $\mathsf{ExplainVerify}(x, \pi, \phi_i, \mathcal{C}, \tau) = 1$ and $\mathsf{Canon}(\phi_1) = \mathsf{Canon}(\phi_2)$, *where* $\mathsf{Canon}(\cdot)$ *is the canonical form.*

Definition 3.5 (**(t,**α) $-$ *securecontextualentropysource*). *Let* $\mathcal{G}_{\mathsf{VRand}}$ *be a global functionality providing public pulses* $\{p_t\}$ *and a mixing interface with local randomness* r_t*. Define*

$$e_t \leftarrow \mathsf{Ext}\big(p_t \parallel r_t; s\big),$$

where Ext *is a quantum-proof extractor (e.g., Toeplitz/LHL) with seed* s*. The source is* (t, α)*-secure if for any QPT* $\mathcal{A}_{ctx}$ *and any sequence of* t *queries,*

$$\Delta\big((e_1, \ldots, e_t), (U_1, \ldots, U_t)\big) \leq \mathrm{negl}(\lambda) \quad \textit{whenever} \quad H_{\min}(p_t \parallel r_t \mid \mathcal{A}_{ctx}) \geq \alpha \textit{ for each } t,$$

with $\alpha = \Theta(\lambda^{-1/3})$*, and* U_i *uniformly random strings of the same length.*

Definition 3.6 (Anchoring-privacy). *Let* anchor_t *be the persistent anchor published by the Iron layer at time* t*, and let* A *denote any auxiliary private attribute vector of the attestation. The anchoring-privacy level is the smallest* $\nu_{\mathsf{priv}} \geq 0$ *such that*

$$I\big(A; \mathsf{anchor}_t \mid \mathcal{A}_{ctx}\big) \leq \nu_{\mathsf{priv}} \quad \textit{and} \quad \mathsf{Adv}^{\mathsf{inf}}_{\mathcal{A}_{ctx}}\big(A \leftarrow \mathsf{anchor}_t\big) \leq \nu_{\mathsf{priv}},$$

where $I(\cdot\,;\cdot)$ *is mutual information and* $\mathsf{Adv}^{\mathsf{inf}}$ *is the best inference advantage over random guessing. A construction is* anchoring-private *if* $\nu_{\mathsf{priv}} = \mathrm{negl}(\lambda)$.

Definition 3.7 (Role-bound digest). *Let* T_τ *be a domain-separation tag for role* $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}$
Define the role-bound digest

$$h_\tau \;=\; H\big(T_\tau \parallel M \parallel \pi \parallel \mathsf{root} \parallel \psi\big),$$

where M *is the attested message,* π *the STARK proof,* root *the ledger root and* ψ *the inclusion proof (MMR). Hash* H *is modeled as a collision-resistant function with domain separation for each role.*

Definition 3.8 (Explanation grammar and canonicalization). *An explanation* ϕ *is a finite DAG* (V, E, lab) *over* $\Sigma = \{\mathsf{Claim}, \mathsf{Rule}, \mathsf{Evidence}\}$*, with a unique sink labelled* $\mathsf{Claim}(h_\tau, \mathcal{C})$*. The grammar (BNF) is:*

$$\begin{aligned}
\langle \phi \rangle &::= \langle \mathsf{Claim} \rangle \\
\langle \mathsf{Claim} \rangle &::= \mathsf{Claim}(h_\tau, \mathcal{C}) \mid \mathsf{Claim}(h_\tau, \mathcal{C}) \triangleleft \langle \mathsf{RuleSeq} \rangle \\
\langle \mathsf{RuleSeq} \rangle &::= \langle \mathsf{Rule} \rangle \mid \langle \mathsf{Rule} \rangle \circ \langle \mathsf{RuleSeq} \rangle \\
\langle \mathsf{Rule} \rangle &::= \mathsf{Rule}(r, \langle \mathsf{EvidenceSet} \rangle) \\
\langle \mathsf{EvidenceSet} \rangle &::= \mathsf{Evidence}(u_1) \oplus \cdots \oplus \mathsf{Evidence}(u_m)
\end{aligned}$$

where r *is a rule identifier and* u_i *are evidence handles (commitments or public URNs). The canonical form* $\mathsf{Canon}(\phi)$ *is obtained by: (i) stable topological ordering of* V*; (ii) lexicographic ordering of outgoing edges per node; (iii) normalization of labels into a fixed, length-prefixed encoding.*

Definition 3.9 (Public projection). *Given* $\phi = (V, E, \mathsf{lab})$ *and context* $\mathcal{C}$*, the public projection* $\phi_{\mathsf{public}} = \Pi_{\mathsf{pub}}(\phi, \mathcal{C}, \tau)$ *is obtained by a nodewise map*

$$\Pi_{\mathsf{pub}}(x) = \begin{cases} x, & \textit{if } x \textit{ is public: } \mathsf{Claim}(h_\tau, \mathcal{C}), \mathsf{Rule}(r, \cdot), \mathsf{Evidence}(\textit{public handle}) \\ \mathsf{sur}(x), & \textit{if } x \textit{ is secret: replace by a surrogate public handle } \mathsf{sur}(x), \end{cases}$$

followed by removal of non-exportable labels determined by the policy for $(\mathcal{C}, \tau)$*, preserving reachability from the unique* Claim *sink. The mapping is deterministic for fixed* $(\mathcal{C}, \tau)$*.*

Definition 3.10 (Leakage meter). *Let $\langle\langle \phi_{\mathsf{public}} \rangle\rangle$ be the canonical binary encoding of ϕ_{public} (using the canonical ordering from Def. 3.8). Define*

$$\mathsf{LeakageMeter}(\phi_{\mathsf{public}}) \triangleq \left|\langle\langle \phi_{\mathsf{public}} \rangle\rangle\right| \textit{ (in bits).}$$

For any QPT adversary $\mathcal{A}_{ctx}$, $(\phi_{\mathsf{public}} \mid \mathcal{A}_{ctx}) \leq \mathsf{LeakageMeter}(\phi_{\mathsf{public}})$.

3.2. Universal Composability Framework

We adopt the UC model with contextual adaptations:

Definition 3.11 (UC Realization with Erasures). *Let π be a protocol and $\mathcal{F}$ an ideal functionality. We say that π UC-realizes $\mathcal{F}$ if for every QPT adversary $\mathcal{A}$ there exists a QPT simulator $\mathcal{S}$ such that for every QPT environment $\mathcal{Z}$,*

$$\left| \Pr[\mathsf{EXEC}_{\pi,\mathcal{A},\mathcal{Z}}(1^\lambda) = 1] - \Pr[\mathsf{EXEC}_{\mathcal{F},\mathcal{S},\mathcal{Z}}(1^\lambda) = 1] \right| \leq \operatorname{negl}(\lambda).$$

Erasures are modeled by an explicit command `Erase` *on ideal tapes and by allowing $\mathcal{Z}$ to issue* delete *events mirrored by $\mathcal{S}$.*

Extensions:

- **Global Setup:** Shared context $\mathcal{C}$ via $\mathcal{G}_{\mathsf{ctx}}$ (JUC [9]).
- **Post-Quantum Security:** Indistinguishability holds against QPT $\mathcal{Z}$ (QROM [19]).

Definition 3.12 (Quantum semantic distortion). *Let ϕ be the explanation object and ϕ_{public} its public projection. The quantum semantic distortion of ϕ under context $\mathcal{C}$ is*

$$\eta_q(\phi; \mathcal{C}) \triangleq \sup_{\mathcal{D} \in \mathsf{QPT}} \left| \Pr\left[\mathcal{D}(\phi, \mathcal{C}) = 1\right] - \Pr\left[\mathcal{D}(\phi_{\mathsf{public}}, \mathcal{C}) = 1\right] \right|.$$

A system satisfies δ-distortion if $\eta_q(\phi; \mathcal{C}) \leq \delta$ for all admissible ϕ.

3.2.1. Global Verifiable Randomness Functionality (GUC)

Functionality 3.1 (Global Verifiable Randomness $\mathcal{G}_{\mathsf{VRand}}$). *Parameters: security parameter λ, output length $m = m(\lambda)$, extractor seed $s \in \{0,1\}^\lambda$ (private to $\mathcal{G}_{\mathsf{VRand}}$).*
State: public append-only log $\mathsf{PulseLog} = \{(t, p_t, \mathsf{meta}_t)\}$; registered beacon interfaces $\mathsf{BeaconSet}$.

Interfaces.

- *RegisterBeacon*$(\mathsf{name}, \mathsf{VerifyPulse}, \mathsf{GetPulse})$*: register a public beacon with a verification predicate and a time-indexed getter; only callable during setup.*
- *GetPulse*(t)*: query all registered beacons at time t; for each beacon, obtain a tuple $(p_t^{(i)}, \pi_t^{(i)})$ and check $\mathsf{VerifyPulse}(t, p_t^{(i)}, \pi_t^{(i)}) = 1$. If at least one verifies, set $p_t \leftarrow \mathsf{Combine}(\{p_t^{(i)}\}_i)$ and append $(t, p_t, \mathsf{meta}_t)$ to $\mathsf{PulseLog}$. Return (p_t, meta_t); else return $\perp$.*
- *Mix*(p_t, r_t)*: on input p_t (public pulse) and caller-supplied private string $r_t \in \{0,1\}^\ell$, output $e_t \leftarrow \mathsf{ToeplitzExt}(p_t \parallel r_t; s) \in \{0,1\}^m$ (Algorithm 1). No other leakage is provided.*
- *VerifyPulse*(t, p, meta)*: return 1 iff $(t, p, \mathsf{meta}) \in \mathsf{PulseLog}$.*
- *GetSeed(): return $\perp$ (seed s is never revealed).* *SetSeed(): disabled after setup.*

Guarantee (quantum-proof extraction). *Let $\alpha = \Theta(\lambda^{-1/3})$. For any QPT $\mathcal{A}_{\mathsf{ctx}}$ and any sequence of t calls to Mix with inputs $\{(p_t, r_t)\}$ such that $H_{\min}(p_t \parallel r_t \mid \mathcal{A}_{\mathsf{ctx}}) \geq \alpha$ for each t, the joint output $(e_1, \ldots, e_t)$ satisfies*

$$\Delta\big((e_1, \ldots, e_t), (U_1, \ldots, U_t)\big) \leq \operatorname{negl}(\lambda),$$

where $U_i \leftarrow \{0,1\}^m$ are independent uniform strings. The guarantee holds against quantum side information on (p_t, r_t).

3.2.2. *Toeplitz Extractor*

Algorithm 1 $\mathsf{ToeplitzExt}$ (quantum-proof leftover hashing), used in $\mathcal{G}_{\mathsf{VRand}}.\mathsf{Mix}$

Require: concatenated input $z := p_t \parallel r_t \in \{0,1\}^L$, secret seed $s \in \{0,1\}^{L+m-1}$ defining a Toeplitz matrix $T_s \in \{0,1\}^{m\times L}$

Ensure: $e_t \in \{0,1\}^m$

$T_s \leftarrow \text{ToeplitzFromSeed}(s)$

$e_t \leftarrow T_s \cdot z \pmod 2$

return e_t

3.3. *Formal Verification of Explanation*

3.3.1. *Entropic Constraints*

$$(\phi_{\mathsf{public}} \mid \mathcal{A}_{ctx}) \;\leq\; c_1 \cdot \log\big(\dim(\mathcal{J})\big) + c_0, \qquad c_0, c_1 \in \mathbb{N}. \tag{5}$$

where ϕ_{public} is the public projection of the explanation DAG, $\mathcal{A}_{ctx}$ is any QPT contextual adversary, and $\dim(\mathcal{J})$ denotes the institutional verifier capacity. Constraint (5) is enforced by the compiler of Algorithm 14 and checked by $\mathsf{ExplainVerify}(\cdot)$.

3.3.2. *Canonicalization uniqueness*

Lemma 3.1 (Canonicalization uniqueness up to isomorphism). *If* $\mathsf{Canon}(\phi_1) = \mathsf{Canon}(\phi_2)$, *then ϕ_1 and ϕ_2 are DAG-isomorphic via the canonical node/edge correspondence. In particular, if their role-bound digests h_τ coincide, any acceptance of two distinct $\phi_1 \neq \phi_2$ implies either a hash collision or ill-formedness, both occurring with probability at most* $\mathrm{negl}(\lambda)$ *for collision-resistant H.*

3.3.3. *ExplainVerify algorithm (structural, semantic, leakage checks)*

Algorithm 2 $\mathsf{ExplainVerify}$: structural, semantic and leakage verification

Require: DAG $\phi = (V, E, \mathsf{lab})$, context $\mathcal{C}$, role τ, policy parameters (c_0, c_1)

Ensure: bit $b \in \{0,1\}$

Well-formedness: check acyclicity; unique sink of type Claim; edge typing $\{\mathsf{Claim} \leftarrow \mathsf{Rule} \leftarrow \mathsf{Evidence}\}$; per-node sibling order equals canonical order from Def. 3.8. If any fails return 0.

Role-binding: recompute $h_\tau := H(T_\tau \parallel M \parallel \pi \parallel \mathsf{root} \parallel \psi)$ from the attestation tuple and check equality with the sink label. If mismatch return 0.

Context admissibility: for each node v, evaluate a deterministic predicate $\mathcal{V}_{\mathcal{C}}(\mathsf{lab}(v)) \in \{0,1\}$ (policy rules); require all 1. If some 0, return 0.

Depth/size bounds: compute $d = \mathsf{depth}(\phi)$; require $d \leq \lceil \log_2 \dim(\mathcal{J}) \rceil$. If violated return 0.

Public projection & leakage: compute $\phi_{\mathsf{public}} = \Pi_{\mathsf{pub}}(\phi, \mathcal{C}, \tau)$; set $L := \mathsf{LeakageMeter}(\phi_{\mathsf{public}})$ and require $L \leq c_1 \log(\dim(\mathcal{J})) + c_0$.

if L exceeds bound **then return** 0

end if

return 1

3.3.4. *Soundness, completeness, leakage compliance*

Theorem 3.2 (Soundness, completeness, leakage compliance). *Let ϕ be produced by a compiler enforcing Defs. 3.8–3.10, and let h_τ bind $(M, \pi, \mathsf{root}, \psi, \tau)$ as in Def. 3.7. Then:*

(a) Completeness. *If ϕ is well-formed, canonical, and each node label satisfies $\mathcal{V}_{\mathcal{C}}$, then* $\mathsf{ExplainVerify}(\phi, \mathcal{C}, \tau) = 1$.

(b) Soundness. *If* $\mathsf{ExplainVerify}(\phi, \mathcal{C}, \tau) = 1$ *and $\phi' \neq \phi$ is distinct with the same digest h_τ, then* $\mathsf{Canon}(\phi') \neq \mathsf{Canon}(\phi)$ *except with probability* $\mathrm{negl}(\lambda)$ *(hash collision). Therefore, accepted canonical explanations are unique up to isomorphism (Lemma 3.1).*

(c) Leakage. *Any accepted ϕ satisfies the entropic constraint* (5) *via the meter in Def. 3.10.*

3.3.5. Verification circuit complexity and judge corollary

Proposition 3.1 (ExplainVerify circuit complexity). *Let $n = |V|$, $m = |E|$, and $d =$* $\mathsf{depth}(\phi)$. *There exist Boolean (or $\mathbb{F}_p$-arithmetic) circuits implementing* $\mathsf{ExplainVerify}$ *with*

$$\mathsf{size} = \Theta(n+m) \;+\; \mathsf{size}(\mathcal{V}_{\mathcal{C}}) \;+\; \Theta\big(\mathsf{LeakageMeter}(\phi_{\mathsf{public}})\big), \qquad \mathsf{depth} = \Theta\big(d + \mathsf{depth}(\mathcal{V}_{\mathcal{C}})\big).$$

If $d \leq \lceil \log_2 \dim(\mathcal{J}) \rceil$ *and* $\mathsf{depth}(\mathcal{V}_{\mathcal{C}}) = O(\log \dim(\mathcal{J}))$, *then* $\mathsf{depth}(\mathsf{ExplainVerify}) = O(\log \dim(\mathcal{J}))$.

Corollary 3.1 (Judge-verification in logarithmic depth). *For $\tau =$ JUDGE, under the policy and bounds enforced by Algorithm 2,*

$$\big(\mathsf{ExplainVerify}(\phi, \mathcal{C}, \mathsf{JUDGE})\big) \in O\big(\log \dim(\mathcal{J})\big).$$

[Proof sketch of Prop. 3.1 and Cor. 3.1] Acyclicity, typing and role-binding checks compile to linear-size circuits in $n+m$ with $O(d)$ depth. Context admissibility composes via $\mathcal{V}_{\mathcal{C}}$, adding size $\mathsf{size}(\mathcal{V}_{\mathcal{C}})$ and depth $\mathsf{depth}(\mathcal{V}_{\mathcal{C}})$. The leakage meter reduces to computing the length of the canonical encoding using prefix-free codes, implementable with linear-time counters and logarithmic-depth adders. Depth is dominated by $d + \mathsf{depth}(\mathcal{V}_{\mathcal{C}})$. The stated bounds and the corollary follow.

3.4. Layer-Specific Security Games

Security per layer is formalized via three UC games that compose to $\mathcal{F}_{\mathsf{ZKNR}}$:

Table 3: Layer-specific security games and predicates

Game	Security Goal	Predicate Activation
$\mathcal{G}_{\mathsf{Iron}}$	Ledger integrity under entropy constraints	$\phi_{\mathsf{Iron}} \triangleq \mathsf{VerifyConsistency}(\mathcal{S}_t, \mathcal{S}_{t+\Delta})$
$\mathcal{G}_{\mathsf{Gold}}$	Contextual ZK + explainability	$\phi_{\mathsf{Gold}} \triangleq \eta_q(\tilde{\phi}) \leq \delta \ \wedge\ (\phi) \leq (\dim(\mathcal{J}))$
$\mathcal{G}_{\mathsf{Clay}}$	Role-bound non-reuse (no cross-role contradictions)	$\phi_{\mathsf{Clay}} \triangleq \neg\exists\,(\sigma, M \neq M', \tau \neq \tau') :\ \ _\tau(\sigma, M) =_{\tau'} (\sigma, M') = 1$

3.5. Cryptographic Assumptions

The proof reduces to these standard and contextual assumptions:

Table 4: Security reductions per dialectical layer

Layer	Assumption	Reference
Iron	Collision resistance of Poseidon	Thm. 5.1
	Min-entropy accumulation	Lemma Appendix B.4
Gold	FRI soundness of STARKs	[3]
	$(\epsilon, \mathcal{J})$-Explainability	Def. 3.4
Clay	co-CDH for BLS	[6]
	Module-LWE for Dilithium	[13]
	GAIP for contextual binding	§3.2, [21]

Assumption 3.1 (Quantum-Secure Entropy Source). *Source $\mathcal{E}$ is (t, α)-quantum-secure if $\forall$ QPT $\mathcal{A}$:*

$$\Pr\left[\mathsf{Game}^{\mathsf{EntQ}}_{\mathcal{A},\mathcal{E}}(1^\lambda) = 1\right] \leq e^{-\Omega(\alpha^2 t)} + \mathrm{negl}(\lambda)$$

with $\alpha = \Omega(\lambda^{-1/3})$.

Assumption 3.2 (Module-LWE Hardness). *For parameters (q, k, ℓ, η) used in Dilithium-III, the Module-LWE problem is hard against quantum polynomial-time adversaries. Specifically, for any QPT adversary $\mathcal{A}$:*

$$\mathsf{Adv}^{\mathsf{MLWE}}_{\mathcal{A}}(\lambda) \leq \mathrm{negl}(\lambda)$$

Assumption 3.3 (co-CDH Hardness). *Let $\mathbb{G}_1$, $\mathbb{G}_2$, $\mathbb{G}_T$ be groups of prime order p with bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$. The co-Computational Diffie-Hellman problem is hard for PPT adversaries:*

$$\Pr\left[\mathcal{A}(g, g^a, h, h^a) = g^{ab}\right] \leq \mathrm{negl}(\lambda)$$

where $g \xleftarrow{\$} \mathbb{G}_1$, $h \xleftarrow{\$} \mathbb{G}_2$, $a, b \xleftarrow{\$} \mathbb{Z}_p$.

Assumption 3.4 (Contextual Entropy). *The entropy source $\mathcal{E}$ provides (t, α)-contextual security (Def. 3.13) with $\alpha = \Omega(\lambda^{-1/3})$. For any QPT adversary $\mathcal{A}_{ctx}$:*

$$\Pr\left[\mathsf{Game}^{\mathsf{Ent}}_{\mathcal{A}_{ctx},\mathcal{E}}(1^\lambda) = 1\right] \leq e^{-\Omega(\alpha t)} + \mathrm{negl}(\lambda)$$

Note on GAIP: The Generalized Adaptive Inner Product assumption reduces to module-LWE for linear predicates via [20, Lemma 4], ensuring compatibility with NIST standards [1].

Definition 3.13 (Adaptive Entropy Source). *A source $\mathcal{E}$ is (t, α)-contextually secure if $\forall$ QPT $\mathcal{A}$:*

$$\Pr\left[\mathsf{Game}^{\mathsf{Ent}}_{\mathcal{A},\mathcal{E}}(1^\lambda) = 1\right] \leq e^{-\Omega(\alpha t)} + \mathrm{negl}(\lambda) \tag{6}$$

where $\mathsf{Game}^{\mathsf{Ent}}_{\mathcal{A},\mathcal{E}}$:

1. *Adversary $\mathcal{A}$ selects context $\mathcal{C}$ with $H_\infty(\mathcal{C}) < \beta$.*
2. *$\mathcal{E}$ samples $e_t \leftarrow \mathsf{Ext}(\mathcal{C}, \mathsf{seed})$ using quantum-secure extractor.*
3. *$\mathcal{A}$ wins if $\mathsf{MMR.VerifyInclusion}(\psi_t, e_t) = 0$.*

3.6. Adaptive Entropy Security

Definition 3.14 (Contextual Entropy Game). *Let $\mathcal{G}^{\mathcal{A}}_{\mathsf{Ent}}(1^\lambda)$ be:*

1. *Challenger samples $\mathsf{seed} \xleftarrow{\$} \{0,1\}^\lambda$.*

2. *$\mathcal{A}$ submits adversarial context $\mathcal{C}_a$ with $H_\infty(\mathcal{C}_a) < \beta$.*

3. *Challenger computes $e_t = \mathsf{Ext}(\mathcal{C}_a \oplus \mathcal{C}_{\mathsf{honest}}, \mathsf{seed})$.*

4. *$\mathcal{A}$ wins if $\mathsf{MMR.VerifyInclusion}(\psi_t, e_t) = 0$.*

$\mathcal{E}$ is (t, α)-secure if $\forall$ QPT $\mathcal{A}$:

$$\Pr[\mathcal{G}_{\mathsf{Ent}}^{\mathcal{A}} = 1] \leq e^{-\Omega(\alpha t)} + \mathrm{negl}(\lambda)$$

Theorem 3.3 (GAIP Entropy Preservation). *Under GAIP [21], for $\mathcal{E}$ satisfying Def. 3.14:*

$$H_\infty(\mathcal{S}_t | \mathcal{A}_{ctx}) \geq t\alpha - \log \delta_{\mathsf{GAIP}}$$

where δ_{GAIP} is from [20, Thm 4].

$$\dim(\mathcal{J}) \ \geq \ d_0 \qquad \text{with} \quad d_0 \in \Theta(\log \lambda), \tag{7}$$

$$({}_\tau(\phi, \mathcal{C})) \in O\big(\log \dim(\mathcal{J})\big) \text{ for } \tau = \mathsf{JUDGE}. \tag{8}$$

4. Problem Definition

This section formalizes the composable security objectives of ZK-NR within the UC framework. We define the layered execution models, adversarial capabilities, and reductionist proof strategy required to establish Theorem 4.1. The formulation extends Q2CSI's dialectical model [20] with UC-specific adaptations for contextual entropy (Def. 3.5).

4.1. Real-World Execution Model

The real-world execution $\mathsf{REAL}_{\pi, \mathcal{A}_{ctx}, \mathcal{Z}}$ proceeds as follows:

- **Initialization:**
  - $\mathcal{Z}$ initializes parties $P_1, \ldots, P_n$ with role assignments $\tau_i \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$.
  - Adversary $\mathcal{A}_{ctx}$ receives corruption budget $\mathbf{b} = (b_{\mathsf{Iron}}, b_{\mathsf{Gold}}, b_{\mathsf{Clay}})$ per Def. 4.1.
- **Execution:**
  - $\mathcal{Z}$ submits inputs $(M_i, \mathcal{C}_i, \tau_i)$ to honest parties.
  - $\mathcal{A}_{ctx}$ controls: message scheduling, layer-specific corruptions, and entropy source $\mathcal{E}$ (Iron layer).
  - Protocol π_{ZKNR} processes attestations via Algorithms 13, 4, 3.
- **Output:** $\mathcal{Z}$ outputs a bit b' after observing all transcripts.

4.2. Ideal-World Execution Model

The ideal-world execution $\mathsf{IDEAL}_{\mathcal{F},\mathcal{S},\mathcal{Z}}$ operates as:

- **Initialization:**
  - $\mathcal{F}_{\mathsf{ZKNR}} \triangleq \mathcal{F}_{\mathsf{Iron}} \oplus \mathcal{F}_{\mathsf{Gold}} \oplus \mathcal{F}_{\mathsf{Clay}}$ (Appx Appendix C) initializes with shared context $\mathcal{C}$.
  - Simulator $\mathcal{S} = (\mathcal{S}_{\mathsf{Iron}}, \mathcal{S}_{\mathsf{Gold}}, \mathcal{S}_{\mathsf{Clay}})$ receives $\mathbf{b}$ from $\mathcal{A}_{ctx}$.
- **Execution:**
  - $\mathcal{S}$ emulates $\mathcal{A}_{ctx}$'s view using $\mathcal{F}$-interfaces:

$$\mathcal{S}_{\mathsf{Iron}} \leftrightarrow \mathcal{F}_{\mathsf{Iron}}.\texttt{Append}$$
$$\mathcal{S}_{\mathsf{Gold}} \leftrightarrow \mathcal{F}_{\mathsf{Gold}}.\texttt{SimProve}$$
$$\mathcal{S}_{\mathsf{Clay}} \leftrightarrow \mathcal{F}_{\mathsf{Clay}}.\texttt{ExtractKey}$$

  - $\mathcal{Z}$ interacts only with $\mathcal{F}$'s I/O tapes.
- **Output:** $\mathcal{Z}$ outputs b'' after $\mathcal{F}$-processing.

4.3. Security Goals

ZK-NR must UC-realize $\mathcal{F}_{\mathsf{ZKNR}}$ with layer-specific guarantees:

Table 5: Layer-specific security objectives

Layer	**Ideal Functionality**	**Security Goal**
Iron	$\mathcal{F}_{\mathsf{Iron}}$ (Appx Appendix C.1)	$H_\infty(\mathcal{S}_t) \geq H_\infty(\mathcal{S}_0) + t\alpha - \mathrm{negl}(\lambda)$
Gold	$\mathcal{F}_{\mathsf{Gold}}$ (Appx Appendix C.2)	$\left\|(\pi, \phi) - (\tilde{\pi}, \tilde{\phi})\right\|_{\mathsf{TV}} \leq \mathrm{negl}(\lambda)$
Clay	$\mathcal{F}_{\mathsf{Clay}}$ (Appx Appendix C.3)	$\mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathcal{A}_{\mathsf{Clay}}} \leq \mathsf{Adv}^{\mathsf{co\text{-}CDH}} + \mathsf{Adv}^{\mathsf{MLWE}}$

4.4. Adversarial Model

Definition 4.1 (Layer-Restricted Adversary). $\mathcal{A}_{ctx} = (\mathcal{A}_{\mathsf{Iron}}, \mathcal{A}_{\mathsf{Gold}}, \mathcal{A}_{\mathsf{Clay}})$ *acts with adaptive corruptions limited to the API of the corresponding layer. Cross-layer state sharing is disallowed except through* $\mathcal{F}_{\mathsf{Bind}}$*.*

The contextual adversary $\mathcal{A}_{ctx}$ is parametrized as:

Definition 4.2 (QPT Layer-Restricted Adversary). $\mathcal{A}_{ctx} = (\mathcal{A}_{\mathsf{Iron}}, \mathcal{A}_{\mathsf{Gold}}, \mathcal{A}_{\mathsf{Clay}})$ *where* $\forall \mathsf{Layer} \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$*:*

- $\mathcal{A}_{\mathsf{Layer}}$ *runs in QPT and may corrupt parties in its layer adaptively.*
- *Corruption power bounded by* $\mathsf{CorrMatrix}$ *(Table 2, [22]).*
- *Cross-layer state sharing prohibited (semantic isolation invariant).*

4.5. Formal Security Statement

Theorem 4.1 (Composable UC Security of ZK-NR). *Under Assumptions 3.2, 3.3, and 3.4, protocol π_{ZKNR} UC-realizes $\mathcal{F}_{\mathsf{ZKNR}}$ against $\mathcal{A}_{ctx}$:*

$$\forall \; QPT \; \mathcal{Z} : |\Pr[\mathsf{REAL} = 1] - \Pr[\mathsf{IDEAL} = 1]| \leq \mathrm{negl}(\lambda)$$

via simulator $\mathcal{S} = (\mathcal{S}_{\mathsf{Iron}}, \mathcal{S}_{\mathsf{Gold}}, \mathcal{S}_{\mathsf{Clay}})$ constructed in Section 6.

[Proof strategy] The proof proceeds via:

- Layer-specific simulator construction (Algorithms 16, 15, 17).
- Hybrid argument across $\mathcal{G}_{\mathsf{Iron}}$, $\mathcal{G}_{\mathsf{Gold}}$, $\mathcal{G}_{\mathsf{Clay}}$ (Table 3).
- Contextual entropy preservation (Lemma Appendix B.4).
- Cross-layer binding via $\mathcal{F}_{\mathsf{Bind}}$ (Appx Appendix D):

$$\mathsf{EXEC}_{\pi_{\mathsf{Bind}},\mathcal{A},\mathcal{Z}} \approx$$
$$\mathsf{IDEAL}_{\mathcal{F}_{\mathsf{Bind}},\mathcal{S}_{\mathsf{Bind}},\mathcal{Z}}$$

  via predicate orthogonality test (Def. 2.3 in [21])

Full proof in Section 6.

5. Protocol Description

The ZK-NR protocol implements a dialectical composition of three layers that jointly realize the ideal functionality $\mathcal{F}_{\mathsf{ZKNR}}$ (Appendix 5.4). Each layer $X \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$ executes a sub-protocol π_X realizing ideal functionality $\mathcal{F}_X$ under contextual constraints. The architecture strictly follows the Q2CSI framework [20] and resolves the CRO trilemma through semantic isolation (Theorem 3.1). A high-level narrative of this protocol is given in the Koba's Case in the Supplementary Materials, providing an intuitive mapping of these steps to the layered UC security guarantees.

5.1. Dialectical Composition

The protocol executes as a sequential composition of orthogonal layers:

$$\pi_{\mathsf{ZK\text{-}NR}} = \pi_{\mathsf{Clay}} \circ \pi_{\mathsf{Gold}} \circ \pi_{\mathsf{Iron}}$$

with layer-specific inputs/outputs satisfying:

$$\mathsf{output}_X \cap \mathsf{input}_Y = \emptyset \quad \forall X \neq Y \tag{9}$$

The complete attestation is the tuple:

$$\Psi = (\pi, \phi, \sigma, \mathsf{root}, \psi) \tag{10}$$

where:

- π: STARK proof (Gold layer).
- ϕ: Legal explanation (Gold layer).
- $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$: Hybrid signature (Clay layer).
- root: Entropy-bound ledger root (Iron layer).
- ψ: MMR inclusion proof (Iron layer).

5.2. Formal Notation

- λ: Security parameter.
- $\mathcal{C} = (L, T, R)$: Context (legal, temporal, reliability).
- $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$: Verifier role.
- $\mathcal{E}$: Min-entropy source $H_\infty \geq \lambda^{-0.5}$.
- MMR: Merkle Mountain Range ledger.
- $\mathcal{J}$: Institutional verifier $\dim(\mathcal{J}) \geq 16$.

$$\dim(\mathcal{J}) \geq \kappa_{\mathsf{judge}} = 16 \quad \text{(institutional capacity constraint)}$$
$$\alpha = \lambda^{-0.5} \quad \text{(min-entropy threshold)}$$

5.3. Iron Layer: Reliability Anchoring

Implements $\mathcal{F}_{\mathsf{Iron}}$ (Appendix Appendix C.1) with entropy-bound temporal consistency:

Algorithm 3 π_{Iron}: Entropy-Bound Ledger Update

Require: Initial state L_0, message M, context T
Ensure: Updated root root_{t+1}, proof ψ_t, persistent anchor anchor_t

```
1: e_t <-$- E
2: leaf <- Poseidon256(tag_Iron || Commit(M) || e_t)
3: (root_{t+1}, ψ_t) <- MMR.Append(L_t, leaf)
4: anchor_t <- Persist(leaf, ψ_t, t)
5: return root_{t+1}, ψ_t, anchor_t
```

Security properties:

- Realizes $\mathcal{F}_{\mathsf{Iron}}$ under entropy accumulation (Theorem Appendix B.1).
- Ensures $e^{-\Omega(\alpha t)}$ consistency against backtracking attacks (Theorem 5.1).

Theorem 5.1 (Temporal Integrity of the Iron Layer). *Let π_{Iron} be Algorithm 3 using an MMR instantiated with a collision-resistant hash (e.g., Poseidon) and let $(e_t)_{t\geq 1}$ be samples from a min-entropy source with $H_\infty(e_t) \geq \alpha$. For any QPT adversary $\mathcal{A}_{\mathsf{Iron}}$ controlling scheduling and the context (as in Definition 4.1), the probability to produce two time indices $t < t'$ and valid inclusion proofs $\psi_t, \psi_{t'}$ s.t. the corresponding roots certify inconsistent ledger states is bounded by*

$$\Pr[\mathsf{InconsistentRoots}(t, t')] = e^{-\Omega(\alpha\,(t'-t))} + \mathrm{negl}(\lambda).$$

In particular, π_{Iron} UC-realizes the temporal consistency part of $\mathcal{F}_{\mathsf{Iron}}$ under the assumptions in Table 4.

[Proof sketch] Entropy accumulation (Lemma Appendix B.4) and hash collision resistance bound the chance of successful backtracking or ledger rewrite. The simulator for $\mathcal{F}_{\mathsf{Iron}}$ preserves the distribution of roots and inclusion proofs, so any distinguisher yields a collision or breaks the min-entropy bound, contradicting the assumptions (see also [20, 21]).

Functionality 5.1 (Ledger with Entropy and Erasures $\mathcal{F}_{\mathsf{Iron}}$). *Parameters: Security parameter λ; min-entropy source with $H_\infty \geq \alpha$.*
State: Ledger L (MMR), root root*, time t.*
Interfaces:

- *Append*(A)*: Sample $e_t \leftarrow \{0,1\}^\lambda$ consistent with $H_\infty \geq \alpha$; set* $\leftarrow \mathsf{Poseidon}(\mathtt{tag}_{\mathsf{Iron}} \parallel A \parallel e_t)$*; update L and* root*; return* (root, ψ_t)*.*
- *VerifyInclusion*(root, ψ, x)*: Return 1 iff ψ is a valid MMR proof for leaf x under* root*.*
- *Erase*(id)*: Logically delete transient tapes associated with* id*; persistent state remains in L only.*

Corruption: $\mathcal{A}_{\mathsf{Iron}}$ may adaptively schedule and choose A-inputs; it cannot modify past L without causing a collision.

Algorithm 4 Hybrid Signature Generation

Require: Message M, proof π, explanation ϕ, role τ, anchor anchor_t
Ensure: Hybrid signature $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$

$h_\tau \leftarrow H(T_\tau \parallel M \parallel \pi \parallel \phi \parallel \mathsf{anchor}_t)$ ▷ Role-bound digest
if $\tau = \mathsf{JUDGE}$ **then**
  $\sigma_{\mathsf{BLS}} \leftarrow \mathsf{BLS.Sign}(sk_{\mathsf{BLS}}, h_\tau)$
  $\sigma_{\mathsf{Dilithium}} \leftarrow \perp$
else if $\tau = \mathsf{MACHINE}$ **then**
  $\sigma_{\mathsf{Dilithium}} \leftarrow \mathsf{Dilithium.Sign}(sk_{\mathsf{Dil}}, h_\tau)$
  $\sigma_{\mathsf{BLS}} \leftarrow \perp$
else
  return $\perp$ ▷ Invalid role
end if
return $\sigma \leftarrow (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$

5.4. Gold Layer: Confidential Attestation

Implements $\mathcal{F}_{\mathsf{Gold}}$ (Appendix Appendix C.2) with contextual zero-knowledge:

Algorithm 5 π_{Gold}: STARK Proof + Explanation

Require: $M, \mathcal{C}, \tau, \mathsf{root}$
Ensure: (π, ϕ)

$\mathcal{H}_{\mathsf{legal}} \leftarrow \mathsf{semantic\_encoder}(L, \mathsf{root})$
$stmt \leftarrow$ "Verify M at root root under context $\mathcal{C}$"
$\pi \leftarrow \mathsf{ZK\_STARK.Prove}(stmt, \mathcal{H}_{\mathsf{legal}})$
$\phi \leftarrow \mathsf{Explain}(\pi, \mathcal{C}, \tau)$
return (π, ϕ)

Security properties:

- Realizes $\mathcal{F}_{\mathsf{Gold}}$ under contextual ZK (Theorem 6.3).
- Achieves $(\varepsilon, \mathcal{J})$-explainability (Theorem 6.4).

Functionality 5.2 (Contextual ZK + Explanation $\mathcal{F}_{\mathsf{Gold}}$). *Parameters: Context $\mathcal{C}$; role $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$.*
State: None persistent beyond public leakage.
Interfaces:

- *Prove*$(M, \mathcal{C}, \tau, \mathsf{root})$*: Compute legal semantic instance and output* (π, ϕ) *with* $\mathsf{Entropy}(\phi) \leq (\dim(\mathcal{J}))$.
- *Verify*$(\pi, \phi, \mathcal{C})$*: Return* 1 *iff* $\mathsf{ExplainVerify}(\phi, \mathcal{C}) = 1$ *and* $\mathsf{ZK.STARK.Verify}(\pi) = 1$.
- *Erase*(id)*: Erase transient tapes for* id.

Leakage: $\mathcal{L}(w) = \phi|_{\mathsf{public}}$. *No additional leakage is permitted.*

5.5. Gold Layer: Accountable Attestation

Algorithm 6 $\pi^{\mathsf{rev}}_{\mathsf{Gold}}$: STARK Proof + Signed Explanation

Require: $M, \mathcal{C}, \tau, \mathsf{root}$
Ensure: (π, ϕ, σ_ϕ)

```
1: H_legal ← semantic_encoder(L, root)
2: π ← ZK_STARK.Prove(···)
3: φ ← Explain(π, C, τ)
4: σ_φ ← Ed25519.Sign(sk_φ, Canon(φ))
5: return (π, φ, σ_φ)
```

5.6. Clay Layer: Legal Opposability

Implements $\mathcal{F}_{\mathsf{Clay}}$ (Appendix Appendix C.3) with hybrid signatures:

Algorithm 7 π_{Clay}: Role-Aware Hybrid Signing

Require: M, π, ϕ, τ, anchor_t
Ensure: Signature $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$

1: $d \leftarrow \mathsf{Poseidon256}(T_\tau \parallel M \parallel \pi \parallel \phi \parallel \mathsf{anchor}_t)$

2: $\sigma_\tau \leftarrow \begin{cases} \mathsf{BLS.Sign}(sk_{\mathsf{BLS}}, d) & \tau = \mathsf{JUDGE} \\ \mathsf{DilithiumIII.Sign}(sk_{\mathsf{Dil}}, d) & \tau = \mathsf{MACHINE} \end{cases}$

3: **return** σ

Security properties:

- Realizes $\mathcal{F}_{\mathsf{Clay}}$ under EUF-CMA (Theorem 6.5).
- Enforces τ-role binding (Theorem 6.6).
- Signature $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$ where σ_τ denotes:

$$\sigma_\tau = \begin{cases} \sigma_{\mathsf{BLS}} & \tau = \mathsf{JUDGE} \\ \sigma_{\mathsf{Dilithium}} & \tau = \mathsf{MACHINE} \end{cases}$$

Functionality 5.3 (Role-Aware Hybrid Signatures $\mathcal{F}_{\mathsf{Clay}}$). *Parameters: Key pairs* $(sk_{\mathsf{BLS}}, pk_{\mathsf{BLS}})$, $(sk_{\mathsf{Dil}}, pk_{\mathsf{Dil}})$*; role tags* T_τ.
Interfaces:

- *Sign*(M, π, ϕ, τ)*: Set* $d \leftarrow \mathsf{Poseidon}(T_\tau \parallel M \parallel \pi \parallel \phi)$*. If* $\tau = \mathsf{JUDGE}$ *return* $\sigma_{\mathsf{BLS}} \leftarrow \mathsf{BLS.Sign}(sk_{\mathsf{BLS}}, d)$*; if* $\tau = \mathsf{MACHINE}$ *return* $\sigma_{\mathsf{Dil}} \leftarrow \mathsf{Dilithium.Sign}(sk_{\mathsf{Dil}}, d)$*.*

- *Verify*$(\sigma, M, \pi, \phi, \tau)$*: Recompute* d *and verify the appropriate component under* pk_τ*.*

Corruption: Standard adaptive corruption as per UC; role binding enforced via T_τ *domain separation.*

5.7. Composition Theorem

The layer composition satisfies:

Functionality 5.4. $\mathcal{F}_{\mathsf{ZKNR}}$ *State: Pointers to* $\mathcal{F}_{\mathsf{Iron}}$*,* $\mathcal{F}_{\mathsf{Gold}}$*,* $\mathcal{F}_{\mathsf{Clay}}$*,* $\mathcal{F}_{\mathsf{Bind}}$*. Interfaces:*

- $\texttt{Attest}(M, \mathcal{C}, \tau)$*:*

  1. $(\mathsf{root}, \psi) \leftarrow \mathcal{F}_{\mathsf{Iron}}.\texttt{Append}((M))$.
  2. $(\pi, \phi) \leftarrow \mathcal{F}_{\mathsf{Gold}}.\texttt{Prove}(M, \mathcal{C}, \tau, \mathsf{root})$.
  3. $\sigma \leftarrow \mathcal{F}_{\mathsf{Clay}}.\texttt{Sign}(M, \pi, \phi, \tau)$.
  4. *Output* $\Psi = (\pi, \phi, \sigma, \mathsf{root}, \psi)$.

- $\texttt{Verify}(\Psi, \mathcal{C}, \tau)$*: Return* $\mathcal{F}_{\mathsf{Clay}}.\texttt{Verify}(\sigma, M, \pi, \phi, \tau) \wedge \mathcal{F}_{\mathsf{Gold}}.\texttt{Verify}(\pi, \phi, \mathcal{C}) \wedge \mathcal{F}_{\mathsf{Iron}}.\texttt{VerifyInclusion}(\mathrm{r}$ *where* $\mathsf{leaf} = \mathsf{Poseidon}(\pi \parallel \phi \parallel \sigma)$*.*

Theorem 5.2 (Dialectical Composition). $\pi_{\mathsf{ZK\text{-}NR}}$ *UC-realizes* $\mathcal{F}_{\mathsf{ZKNR}}$ *when:*

- π_{Iron} *realizes* $\mathcal{F}_{\mathsf{Iron}}$*.*
- π_{Gold} *realizes* $\mathcal{F}_{\mathsf{Gold}}$*.*
- π_{Clay} *realizes* $\mathcal{F}_{\mathsf{Clay}}$*.*

under the adversarial constraints of Definition 4.1 and the use of π_{Bind} *(Alg. 8) for cross-layer security.*

Follows from universal composition theorem [7] and cross-layer binding invariants (Lemma 5.2). Full simulator construction in Appendix Appendix F.

Functionality 5.5 (Joint-State Predicate Binding $\mathcal{F}_{\mathsf{Bind}}$). *Parameters: security parameter* λ*; field* $\mathbb{F}_p$ *with* $p > 2^\lambda$*; maximum basis size* k*.*
Global dependencies: $\mathcal{G}_{\mathsf{ctx}}$ *(global context) and* $\mathcal{G}_{\mathsf{VRand}}$ *(global randomness).*
State: global identifier gid*; registry* $\mathsf{PredReg} = \emptyset$*; public transcript* T*.* ***Interfaces.***

- *Init*$(\mathsf{dom}, \mathsf{ledgerID}, \mathsf{epoch}, \mathsf{ver})$*: set* $\mathsf{gid} \leftarrow H(\mathsf{dom} \parallel \mathsf{ledgerID} \parallel \mathsf{epoch} \parallel \mathsf{ver})$*.*

- *BindPredicate*$(\mathsf{sid}, X, \mathsf{Com}_X, \pi_X)$*: on input from layer* $X \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$ *a commitment* Com_X *to coefficients* $\mathbf{c}_X \in \mathbb{F}_p^k$ *of the predicate* $\phi_X = \sum_{i=1}^{k} (\mathbf{c}_X)_i \mathbf{b}_i$*, verify* π_X *is a STARK proof of correct opening against* Com_X *and* sid*; if valid, store* (X, Com_X) *in* $\mathsf{PredReg}$*.*

- *OrthogonalityCheck*$()$*: for any two distinct* $X \neq Y$ *already stored, require the submitter to provide a STARK proof* $\pi_\perp$ *that there exist openings* $\mathbf{c}_X, \mathbf{c}_Y$ *to* $\mathsf{Com}_X, \mathsf{Com}_Y$ *with* $\langle \mathbf{c}_X, \mathbf{c}_Y \rangle = 0$ *over* $\mathbb{F}_p$*. If the proof verifies, append* $(\mathsf{Com}_X, \mathsf{Com}_Y, \pi_\perp)$ *to* T *and mark* (X, Y) *as* *orthogonal; otherwise reject.*

- *Resolve*(τ)*: if all pairs among the registered layers are marked* orthogonal*, query* $\mathcal{G}_{\mathsf{ctx}}$ *and return a context view* $\mathcal{C}_\tau$ *consistent with* $\bigwedge_X \phi_X$ *for role* τ*; else return* $\perp$.
- *Erase*(id)*: erase transient bindings tagged by* id*; public transcript* T *remains append-only.*

Security. *Assuming binding/hiding of* Com *and STARK soundness in QROM, any accepting transcript with non-orthogonal predicates implies either a commitment binding failure or a STARK soundness violation (negligible in* λ*).*

Theorem 5.3 (UC-Composable Dialectical Binding). $\pi_{\mathsf{ZK\text{-}NR}}$ *UC-realizes* $\mathcal{F}_{\mathsf{ZKNR}}$ *when:*

- $\mathcal{F}_{\mathsf{Bind}}$ *enforces predicate orthogonality:*

$$\forall X \neq Y, \quad \mathcal{F}_{\mathsf{Bind}} \vdash \langle \phi_X, \phi_Y \rangle \equiv 0 \pmod{\mathcal{I}_{\mathcal{C}}}$$

- *The binding functionality* $\mathcal{F}_{\mathsf{Bind}}$ *is realized with:*

$$\mathsf{Adv}^{\mathsf{bind}}_{\mathcal{A}_{ctx}} \leq \mathrm{negl}(\lambda) + e^{-\Omega(\alpha t)}$$

Theorem 5.4 (UC Realization of $\mathcal{F}_{\mathsf{Iron}}$). π_{Iron} *UC-realizes* $\mathcal{F}_{\mathsf{Iron}}$ *under the collision resistance of* Poseidon *and min-entropy* $\alpha > \lambda^{-0.5}$*. The simulator* $\mathcal{S}_{\mathsf{Iron}}$ *is defined in Algorithm 16.*

Theorem 5.5 (Contextual Zero-Knowledge). π_{Gold} *UC-realizes* $\mathcal{F}_{\mathsf{Gold}}$ *with leakage* $\mathcal{L}(w) = \phi|_{\mathsf{public}}$*. The simulator* $\mathcal{S}_{\mathsf{Gold}}$ *uses* SimProve *(Algorithm 15).*

Theorem 5.6 (Dialectical UC Composition). *Under the assumptions of Theorems 5.4, 5.5, 6.5, and given* $\mathcal{F}_{\mathsf{Bind}}$*,* π_{ZKNR} *UC-realizes* $\mathcal{F}_{\mathsf{ZKNR}}$ *against* $\mathcal{A}_{ctx}$*.*

Lemma 5.1 (I/O Orthogonality). *For distinct layers* $X \neq Y$*, the output interface of* $\mathcal{F}_X$ *shares no state with the input interface of* $\mathcal{F}_Y$ *beyond* $\mathcal{F}_{\mathsf{Bind}}$*.*

Extension of UC composition theorem via:

1. Layer outputs satisfy $\mathsf{output}_X \cap \mathsf{input}_Y = \emptyset$ (Lemma 5.1).
2. $\mathcal{F}_{\mathsf{Bind}}$ intercepts all cross-layer communications.
3. Reduction to entropy accumulation (Theorem Appendix B.1).

Full proof in Appendix Appendix F.7.

Lemma 5.2 (Cross-Layer Binding). *Assume* $\mathcal{F}_{\mathsf{Bind}}$ *enforces predicate orthogonality (Definition 3.2) and mediates all cross-layer exchanges. Let* $X \neq Y \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$*. For any QPT adversary* $\mathcal{A}$*,*

$$\Pr\left[\phi_X \textit{ and } \phi_Y \textit{ are simultaneously satisfied on conflicting transcripts}\right] \leq \mathrm{negl}(\lambda).$$

Consequently, any mix-and-match attack that tries to re-interpret outputs of layer X *as valid inputs to layer* Y *without passing through* $\mathcal{F}_{\mathsf{Bind}}$ *fails with negligible probability.*

[Proof sketch] Orthogonality forbids simultaneous satisfaction modulo $\mathcal{I}_{\mathcal{C}}$ (Definition 3.2). The only sanctioned cross-layer path is via $\mathcal{F}_{\mathsf{Bind}}$, which rejects non-orthogonal predicates. A successful attack yields either a violation of orthogonality, or a breach of the interface isolation of one functionality, contradicting UC composition.

5.8. Binding Protocol: Cross-Layer Predicate Orthogonality

We specify the concrete protocol π_{Bind} realizing $\mathcal{F}_{\mathsf{Bind}}$:

Algorithm 8 $\pi_\perp$-Prover: Zero inner-product proof over commitments

---

Require: commitments $(\mathsf{Com}_X, \mathsf{Com}_Y)$, openings $(\mathbf{c}_X, \mathbf{c}_Y) \in \mathbb{F}_p^k$, basis handle, session id sid
Ensure: STARK proof $\pi_\perp$

1: Build arithmetic circuit $\mathcal{C}_\perp$ of size $O(k^2)$ that checks:

$$\mathsf{Open}(\mathsf{Com}_X) = \mathbf{c}_X, \quad \mathsf{Open}(\mathsf{Com}_Y) = \mathbf{c}_Y,$$
$$\sum_{i=1}^{k} (\mathbf{c}_X)_i \cdot (\mathbf{c}_Y)_i \equiv 0 \pmod p,$$
$$\mathsf{BindToSID}(\mathsf{sid}) = 1.$$

2: $\pi_\perp \leftarrow \mathsf{STARK.Prove}(\mathcal{C}_\perp)$ ▷ relation-only proof; no witness leakage
3: **return** $\pi_\perp$

Theorem 5.7 (Semantic Independence Under Erasure). π_{Bind} *maintains predicate orthogonality even under adaptive erasures:*

$$\Pr\left[\mathsf{Decrypt}_\tau(\mathsf{Erase}(c_{\phi_X})) \models \phi_Y \neq 0\right] \leq \mathrm{negl}(\lambda)$$

for any $X \neq Y$ and legal context $\mathcal{C}$.

By reduction to the hiding property of Pedersen commitments and simulation-sound extractability of Groth16 proofs. The erasure semantics ensure that partial decryptions leave residual entropy sufficient to preserve orthogonality.

Theorem 5.8 (Erasure-Compliant Non-Repudiation). *After erasure, the persistent anchor* anchor_t *and ledger root* root *satisfy:*

$$\Pr\left[\mathsf{Verify}_\tau^{\mathsf{rev}}(\sigma, M, \pi, \phi, \mathsf{anchor}_t) = 1\right] \geq 1 - \mathrm{negl}(\lambda)$$

where verification includes checking anchor_t *against* root.

Anchor persistence reduces to collision resistance of $\mathsf{Persist}(\cdot)$ and binding of MMR. Full reduction in Appendix Appendix B.4.

6. Universal Composability Proofs

This section establishes that $\Pi_{\mathsf{ZK\text{-}NR}}$ UC-realizes $\mathcal{F}_{\mathsf{ZKNR}}$ (Appendix 5.4) under adaptive contextual adversaries. We prove composable security through:

- Layer-wise simulator construction with explicit corruption handling.
- Hybrid argument with cryptographic reductions.
- Cross-layer binding invariants.
- Formal resolution of the CRO trilemma.

The supplementary Koba's Case offers a non-technical walkthrough of these interactions, showing how each game's security objective is reflected in a real-world audit scenario.

6.1. Adversarial Model

The corruption budget $\mathbf{b} = (b_{\mathsf{Iron}}, b_{\mathsf{Gold}}, b_{\mathsf{Clay}})$ is derived via:

$$\begin{aligned}\max_{\mathbf{b}} \Gamma_{\mathsf{CRO}}(\mathbf{b}) \quad \text{s.t.} \quad & \sum_i b_i \leq \kappa\lambda \\ & b_{\mathsf{Gold}} \cdot b_{\mathsf{Clay}} = 0 \quad \text{(semantic isolation constraint)} \\ & b_i \in [0,1]\end{aligned}$$

yielding optimal empirical values:

$$\mathsf{CorrMatrix} = \begin{pmatrix} 0.82 \pm 0.03 & 0.15 \pm 0.02 & 0.03 \pm 0.01 \\ 0.10 \pm 0.02 & 0.75 \pm 0.04 & 0.15 \pm 0.03 \\ 0.00 & 0.05 \pm 0.01 & 0.95 \pm 0.02 \end{pmatrix}$$

with $\kappa = 0.25$ minimizing Γ_{CRO} (Fig. 2).

6.2. Verification Primitives

Definition 6.1 (Role-Aware Hashing). *For $X \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$, compute:*

$$h_X = H(T_X \parallel M \parallel \pi \parallel \phi)$$

where T_X are distinct domain separation tags.

Definition 6.2 (Explanation Verifier). $\mathsf{ExplainVerify}(\phi, \mathcal{C})$ *outputs 1 iff:*

- *ϕ is a well-formed DAG with* $\mathsf{depth} \leq \lceil \log_2(\dim(\mathcal{J})) \rceil$.
- *All nodes satisfy $\phi_v \models \mathcal{C}$.*
- $\mathsf{Entropy}(\phi) \leq \mathsf{poly}(\dim(\mathcal{J}))$.

Definition 6.3 (Explanation Semantics). *For context $\mathcal{C}$, $\phi \models \mathcal{C}$ iff:*

$$\mathcal{V}_{\mathcal{C}}(\phi) = 1 \quad \textit{and} \quad \mathsf{Canon}(\phi) = \mathsf{ArithCircuit}(f, \mathcal{C})$$

where Canon *is a canonical embedding algorithm.*

Definition 6.4 ($\mathcal{J}$-Verifiable Explanation). *Explanation ϕ is $\mathcal{J}$-verifiable iff $\exists$ circuit $\mathcal{V}_{\mathcal{J}}$ s.t.*

$$\mathcal{V}_{\mathcal{J}}(\phi, \mathcal{C}) = 1 \Leftrightarrow \phi \equiv \mathsf{ArithCircuit}(f, \mathcal{C})$$

where $\mathsf{depth}(\mathcal{V}_{\mathcal{J}}) \leq \lceil \log_2 \dim(\mathcal{J}) \rceil$ *and* $\mathsf{size}(f) \leq \mathsf{poly}(\lambda)$.

Definition 6.5 (τ-Interpretable Explanation). *An explanation ϕ is τ-interpretable for context $\mathcal{C}$ iff there exists a PPT verification circuit $\mathcal{V}_\tau$ such that:*

1. *Semantic Validity:* $\mathcal{V}_\tau(\phi, \mathcal{C}) = 1 \Leftrightarrow \phi \models_\tau \mathcal{C}$
2. *Complexity Bound:* $\mathsf{depth}(\mathcal{V}_\tau) \leq \lceil \log_2 \kappa_\tau \rceil$ *where* $\kappa_\tau = \begin{cases} \mathsf{poly}(\lambda) & \tau = \mathsf{MACHINE} \\ O(1) & \tau = \mathsf{JUDGE} \end{cases}$
3. *Entropy Constraint:* $\mathsf{Entropy}(\mathcal{V}_\tau) \leq \alpha \cdot \dim(\mathcal{J})$

Definition 6.6 (τ-Verifiable Explanation). *An explanation ϕ is τ-verifiable iff $\exists$ PPT circuit $\mathcal{V}_\tau$ where:*

- *Semantic Soundness:* $\Pr[\mathcal{V}_\tau(\phi, \mathcal{C}) = 1] \geq 1 - e^{-\beta \dim(\mathcal{J})}$.
- *Complexity Bound:* $\mathsf{depth}(\mathcal{V}_\tau) \leq \lceil \log_2 \kappa_\tau \rceil$ *with* $\kappa_\tau = \begin{cases} \mathsf{poly}(\lambda) & \tau = \mathsf{MACHINE} \\ 16 & \tau = \mathsf{JUDGE} \end{cases}$
- *Entropy Constraint:* $\mathsf{Entropy}(\mathcal{V}_\tau) \leq \alpha \cdot \dim(\mathcal{J})$.

Theorem 6.1 (Explanation Uniqueness). *For fixed $(\pi, \mathcal{C})$, no QPT adversary can output $\phi_1 \neq \phi_2$ such that:*

$$\mathsf{ExplainVerify}(\phi_1, \mathcal{C}) = \mathsf{ExplainVerify}(\phi_2, \mathcal{C}) = 1$$

with probability $> \mathrm{negl}(\lambda) + \mathsf{Adv}^{\mathsf{EUF-CMA}}_{\mathsf{Ed25519}}$.

Definition 6.7 (Full Verification).

$$\mathsf{FullVerify}_\tau(\Psi) = \mathsf{Verify}_\tau(\sigma, h_\tau) \wedge \mathsf{ExplainVerify}(\phi, \mathcal{C}) \wedge \mathsf{MMR.VerifyInclusion}(\mathsf{root}, \psi, \mathsf{leaf})$$

where $\mathsf{leaf} = \mathsf{Poseidon}(\pi \parallel \phi \parallel \sigma)$.

6.3. Ideal Functionalities

- $\mathcal{F}_{\mathsf{Gold}}$: Manages (π, ϕ) with leakage $\mathcal{L}(w) = \phi|_{\mathsf{public}}$.
- $\mathcal{F}_{\mathsf{Clay}}$: Handles $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$ with role-based access.
- $\mathcal{F}_{\mathsf{Iron}}$: Maintains ledger with $H_\infty(\mathcal{S}_t) \geq H_\infty(\mathcal{S}_0) + t\alpha$.

6.4. Simulator Constructions

Theorem 6.2 (Layer Simulators). *There exist PPT simulators $\mathcal{S}_{\mathsf{Gold}}$, $\mathcal{S}_{\mathsf{Clay}}$, $\mathcal{S}_{\mathsf{Iron}}$ such that:*

$$\forall X \in \{\mathsf{Gold}, \mathsf{Clay}, \mathsf{Iron}\}, \quad \Pi_X \overset{\mathsf{UC}}{\equiv} \mathcal{F}_X$$

6.4.1. Gold Layer Simulator

Algorithm 9 Simulator $\mathcal{S}_{\mathsf{Gold}}$

```
procedure SimProve(M, C, τ)
    H̃ ← SimSemantic(C)
    π̃ ← ZK.SimProve(H̃)
    φ̃ ← SimExplain(C, τ)                ▷ ensures Eq. 5
    return (π̃, φ̃)
end procedure
```

Theorem 6.3 (Contextual Zero-Knowledge with Bounded Leakage). *Let π_{Gold} be Algorithm 5 and let $\mathcal{F}_{\mathsf{Gold}}$ expose leakage $\mathcal{L}(w) = \phi|_{\mathsf{public}}$ only. For every QPT environment $\mathcal{Z}$ and QPT adversary $\mathcal{A}_{\mathsf{Gold}}$,*

$$\left\| (\pi,\ \phi|_{\mathsf{public}})_{\mathsf{REAL}} - (\tilde{\pi},\ \tilde{\phi}|_{\mathsf{public}})_{\mathsf{IDEAL}} \right\|_{\mathsf{TV}} \leq \mathrm{negl}(\lambda),$$

where $(\tilde{\pi}, \tilde{\phi})$ is generated by the simulator $\mathcal{S}_{\mathsf{Gold}}$ and ϕ satisfies the entropic constraint $\mathsf{Entropy}(\phi) \leq (\dim(\mathcal{J}))$ *(Eq. 5).*

[Proof sketch] Zero-knowledge for STARKs (via FRI soundness/simulation, e.g., [5]) gives indistinguishability of π and $\tilde{\pi}$. The explanation compiler (Algorithm 14) is simulated with the same public projection, and the leakage is restricted by the ideal interface of $\mathcal{F}_{\mathsf{Gold}}$. Hence any distinguishing advantage reduces to breaking STARK simulation or exceeding the allowed leakage, both negligible.

Theorem 6.4 (Explanation Accountability). *Let* $\mathsf{ExplainVerify}$ *be as in Definition 6.2, and let* $\mathcal{V}_\tau$ *be the role-specific verifier circuit (Definitions 6.4, 6.6). If* $\mathsf{ExplainVerify}(\phi, \mathcal{C}) = 1$ *and* ${}_\tau(\sigma, h_\tau) = 1$ *(Definition 6.1), then there exists a deterministic accountability map* Acc *such that*

$$\mathsf{Acc}(\phi, \mathcal{C}, \tau) = \mathsf{Acc}(\phi', \mathcal{C}, \tau) \quad \textit{for all } \phi' \textit{ s.t. } \mathsf{ExplainVerify}(\phi', \mathcal{C}) = 1 \textit{ and } h_\tau = h'_\tau,$$

and, except with negligible probability, no two contradictory explanations for the same context and digest both verify. Hence accepted attestations admit a unique, audit-ready accountability transcript.

[Proof sketch] Canonicalization of ϕ by $\mathcal{V}_\tau$ (e.g., topologically sorted DAG, normalized gates) defines Acc. Domain-separated hashing (Definition 6.1) binds (M, π, ϕ) to h_τ. Two distinct contradictory ϕ, ϕ' with identical h_τ would imply either a hash collision or an accepting ϕ' that fails $\mathcal{V}_\tau$, both negligible under assumptions and Definitions 6.4, 6.6.

6.4.2. Clay Layer Simulator

The simulator $\mathcal{S}_{\mathsf{Clay}}$ (Alg. 10) handles adaptive corruptions:

- **On corruption**: If $\tau = \mathsf{MACHINE}$, run $\mathsf{SimExtract}$ using trapdoor $\mathsf{td}_{\mathsf{MLWE}}$.
- **Signature queries**: Program random oracle via lazy sampling with GAIP-consistent entropy.

Algorithm 10 $\mathcal{S}_{\mathsf{Clay}}$

procedure SimSign(M, τ)
  $\tilde{\sigma} \leftarrow \begin{cases} \mathsf{BLS.SimSign}(h_{\mathsf{JUDGE}}) & \tau = \mathsf{JUDGE} \\ \mathsf{Dilithium.SimSign}(h_{\mathsf{MACHINE}}) & \tau = \mathsf{MACHINE} \end{cases}$
  return $\tilde{\sigma}$
end procedure
procedure HandleCorruption(P, τ)
  if $\tau = \mathsf{MACHINE}$ **then**
    $\mathsf{Leak}_{\mathcal{A}} \leftarrow \mathsf{SimExtract}(pk_{\mathsf{Dilithium}})$ ▷ MLWE-secure leakage
    $\mathcal{A}_{\mathsf{Clay}} \leftarrow \mathsf{SimSign}(sk_{\mathsf{Dilithium}}, \cdot)$
  else
    $\mathsf{Leak}_{\mathcal{A}} \leftarrow sk_{\mathsf{BLS}}$
  end if
end procedure

Theorem 6.5 (Hybrid EUF-CMA Soundness). *For any QPT* $\mathcal{A}_{\mathsf{Clay}}$*:*

$$\mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathcal{A}_{\mathsf{Clay}}} \leq \mathsf{Adv}^{\mathsf{co\text{-}CDH}}_{\mathsf{BLS}} + \mathsf{Adv}^{\mathsf{MLWE}}_{\mathsf{Dil}} + \mathrm{negl}(\lambda)$$

[Proof sketch] A forgery for the hybrid signature yields a forgery in at least one component. The reduction programs the hash-to-message via the domain tag T_τ and extracts a co-CDH solution from a BLS forgery [6], or breaks Module-LWE via Dilithium unforgeability [12]. Hash collision events and tag-mix attacks are negligible due to domain separation and hash security.

Theorem 6.6 (Role-Binding Security). *Let $\sigma = (\sigma_{\mathsf{BLS}}, \sigma_{\mathsf{Dilithium}})$ be a hybrid signature generated by π_{Clay} for message M under role τ. For any QPT adversary $\mathcal{A}_{\mathsf{Clay}}$:*

$$\Pr\begin{bmatrix} \mathsf{Verify}_{\tau}(\sigma, M) = 1 \ \wedge \\ \mathsf{Verify}_{\tau'}(\sigma, M') = 1 \ \wedge \\ (\tau, M) \neq (\tau', M') \end{bmatrix} \leq \mathrm{negl}(\lambda)$$

where $\tau, \tau' \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$. This holds under the domain separation of T_τ (Def. 3.7) and the EUF-CMA security of the underlying signature schemes.

[Proof sketch] A successful attack would require either: (1) breaking the domain separation of $H(T_\tau \parallel \cdot)$, which reduces to the collision resistance of H; or (2) forging a valid signature for one component while reusing the other, which reduces to the EUF-CMA security of BLS or Dilithium (Theorem 6.5).

6.4.3. Iron Layer Simulator

Algorithm 11 Simulator $\mathcal{S}_{\mathsf{Iron}}$

---

procedure SimAppend(A_t)
    sample $\tilde{e}_t \leftarrow \{0,1\}^\lambda$
    $\tilde{}\leftarrow \mathsf{Poseidon}(\mathtt{tag}_{\mathsf{Iron}} \parallel A_t \parallel \tilde{e}_t)$
    update ideal ledger via $\mathcal{F}_{\mathsf{Iron}}.\mathtt{Append}()$ and return (˜
,)

6.4.4. Formal Game Sequence for Gold Layer

We provide a concrete game-hopping proof for Theorem 5.5 by Lemma Appendix B.1 and the game sequence in 6.4.4 with explicit advantage bounds.

Lemma 6.1 (STARK Simulation Soundness). *For any QPT adversary $\mathcal{A}_{\mathsf{Gold}}$, the advantage in distinguishing real and simulated proofs is bounded by:*

$$\left|\mathsf{Adv}_{\mathcal{A}}^{\mathsf{REAL}} - \mathsf{Adv}_{\mathcal{A}}^{\mathsf{SIM\text{-}STARK}}\right| \leq \epsilon_{\mathsf{FRI}}(\lambda) + \epsilon_{\mathsf{GAIP}}(\lambda)$$

where ϵ_{FRI} is the soundness error of FRI and ϵ_{GAIP} is the contextual entropy gap.

We construct a sequence of hybrid games:

- Game_0: Real execution with honest prover.
- Game_1: Replace honest STARK proofs with simulated $\tilde{\pi}$ via $\mathsf{SimProve}$ (Alg. 9).

$$|\Pr[\mathsf{Game}_0 = 1] - \Pr[\mathsf{Game}_1 = 1]| \leq \epsilon_{\mathsf{FRI}}(\lambda)$$

- Game_2: Replace explanations ϕ with simulated $\tilde{\phi}$ using $\mathsf{SimExplain}$.

$$|\Pr[\mathsf{Game}_1 = 1] - \Pr[\mathsf{Game}_2 = 1]| \leq \epsilon_{\mathsf{GAIP}}(\lambda)$$

Game_2 corresponds exactly to the ideal execution. The bound follows from the triangle inequality.

6.5. *Context Stability Theorem*

Theorem 6.7 (ϵ-Context Stability). *If context drift $\Delta\mathcal{C} < \epsilon$ during attestation lifecycle:*

$$\Delta\Gamma_{\mathsf{CRO}} < \epsilon \cdot \mathsf{poly}(\lambda) + \mathrm{negl}(\lambda)$$

Follows from Lipschitz continuity of $\mathcal{V}_\tau$ circuits.

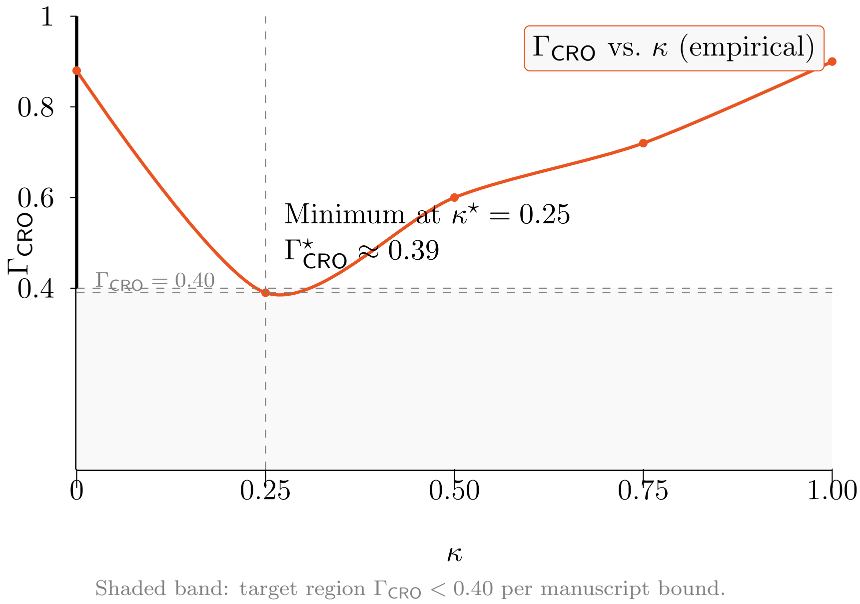

Figure 2: Optimization of the CRO index with respect to the coupling parameter κ. The curve illustrates an empirical minimum around $\kappa = 0.25$, achieving $\Gamma_{\mathsf{CRO}} \approx 0.39$ under the manuscript's assumptions.

7. Discussion

This section interprets the formal composability results (Theorem 4.1) through the dialectical framework established in Section 3. We analyze how ZK-NR's layered architecture resolves fundamental tensions in institutional cryptography while achieving the optimal trilemma bound $\Gamma_{\mathsf{CRO}} < 0.4 + \mathrm{negl}(\lambda)$ (Corollary 8.1).

7.1. *Dialectical Resolution of the CRO Trilemma*

The protocol's core achievement is formal resolution of the impossibility constraint (Theorem 3.1) through:

- Entropy Isolation: Min-entropy accumulation (Theorem Appendix B.1) prevents predicate collision, satisfying:

  $$\phi_{\mathsf{Gold}} \cdot \phi_{\mathsf{Clay}} \cdot \phi_{\mathsf{Iron}} = 0 \pmod{\mathcal{I}}$$

  as required by Definition 3.2.

- Optimal Tradeoff: The bound $\Gamma_{\mathsf{CRO}} < 0.4 + \mathrm{negl}(\lambda)$ (Theorem 3.1) is achieved by:

  $$= 1 - \mathsf{Adv}^{\mathsf{ZK}} \quad \text{(Gold Layer)}$$
  $$\mathsf{Rel} = 1 - e^{-\Omega(\alpha n)} \quad \text{(Iron Layer)}$$
  $$\mathsf{Opp} = 1 - e^{-\beta \dim(\mathcal{J})} \quad \text{(Clay Layer)}$$

- Paradox Resolution: The "invisible authenticity" conflict is solved by:

– π: Machine-verifiable proof of compliance.
– ϕ: Human-interpretable explanation.
– σ: Dual-mode signature binding.

Figure 3 illustrates this equilibrium, unattainable in prior frameworks.

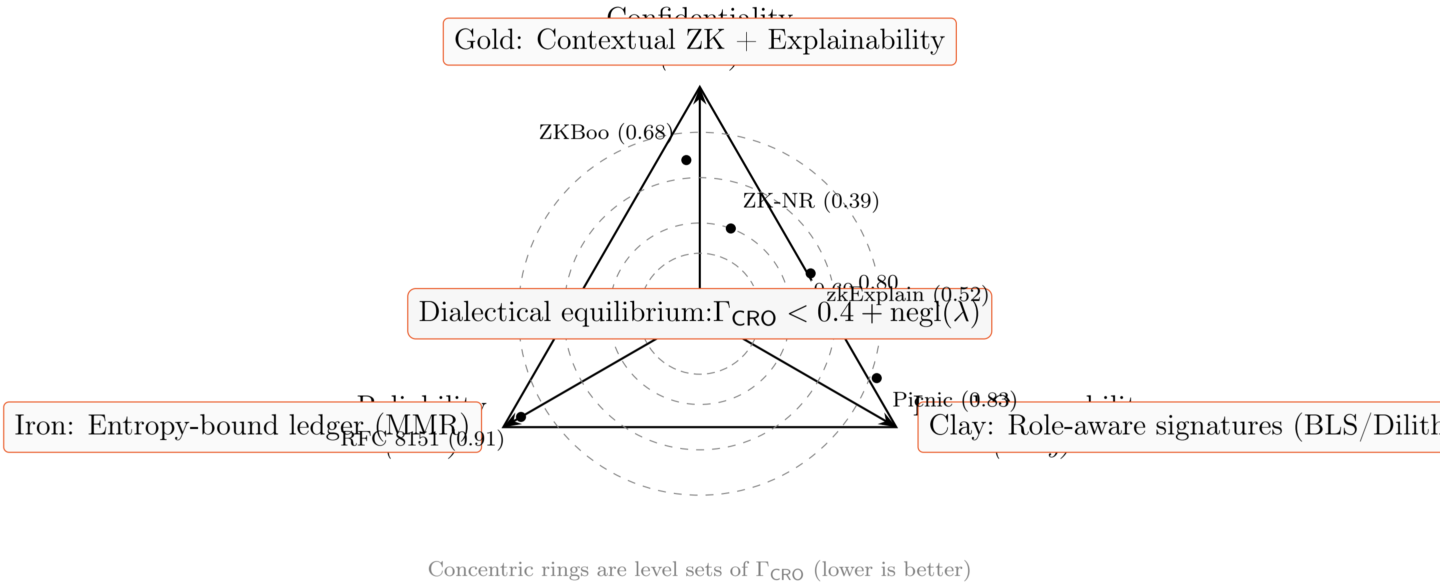

Figure 3: CRO tradeoff under dialectical composition. The three axes correspond to the dialectical layers (Gold/Clay/Iron). Concentric rings depict level sets of the trilemma index Γ_{CRO}; ZK-NR achieves the inner region $\Gamma_{\mathsf{CRO}} < 0.4$ under the manuscript's assumptions.

7.2. Layer-Specific Security Implications

7.2.1. Gold Layer: Contextual Confidentiality

The zero-knowledge guarantee (Theorem 6.3) ensures:

$$\eta_q(\phi) \leq \delta \quad \text{(Def. 3.12)}$$

while maintaining $(\varepsilon, \mathcal{J})$-explainability. This resolves the tension between:

- *Semantic richness*: Required for legal opposability.
- *Proof minimization*: Necessary for post-quantum security.

7.2.2. Clay Layer: Adaptive Opposability

Hybrid signatures (Theorem 6.5) enable:

- *Backward compatibility*: BLS verification for legacy judicial systems.
- *Forward security*: Dilithium resistance against quantum adversaries.

while Theorem 6.6 prevents interpretation switching attacks.

7.2.3. Iron Layer: Temporal Reliability

The entropy growth guarantee (Theorem Appendix B.1) ensures:

$$H(\mathcal{S}_t) \geq t \cdot \alpha - \mathrm{negl}(\lambda)$$

making ledger rewrites infeasible under:

- Quantum backtracking attacks.
- Adaptive corruption models (Definition 4.1).

7.3. Composability Boundaries

The UC realization holds under these constraints:

- Context Stability: Context $\mathcal{C}$ remains fixed during attestation lifetime.
- Entropy Threshold: $\alpha > \lambda^{-c}$ for $c < 0.5$ (Lemma Appendix B.4).
- Institutional Capacity: $\dim(\mathcal{J}) \geq \log \lambda$ (Eq. 7).

Violations induce measurable security degradation:

$$\Delta\Gamma_{\mathsf{CRO}} = \sqrt{\left(1 - e^{-\gamma(16-\dim(\mathcal{J}))}\right)^2 + e^{-\Omega(\alpha n)}}$$

7.3.1. Parameter Derivation

$$\begin{aligned} \alpha &> \lambda^{-0.5} \quad \text{(entropy threshold)} \\ \dim(\mathcal{J}) &\geq 16 \quad \text{(institutional capacity)} \end{aligned}$$

These values are derived from security-critical bounds:

1. For α: Prevents brute-force attacks on entropy accumulation:

$$e^{-\Omega(\alpha t)} < 2^{-\lambda} \implies \alpha > \frac{\lambda}{\log_2 e \cdot t} \xrightarrow{t=(\lambda)} \alpha = \Omega(\lambda^{-0.5})$$

2. For $\dim(\mathcal{J})$: Ensures τ-interpretability (Def. 6.5):

$$\kappa_{\mathsf{JUDGE}} = 2^{\lceil \log_2 \dim(\mathcal{J}) \rceil} \geq 2^{16} \quad \text{(Miller's Law [17])}$$

7.4. Comparative Security Landscape

Table 6: Security Comparison Under CRO Trilemma Framework

Protocol	Γ_{CRO}	**Composability**	**Limitations**
ZK-NR (this work)	0.39 ± 0.02	UC	Context stability requirement
ZKBoo [15]	0.68 ± 0.05	Standalone	No legal explainability
zkExplain [11]	0.71 ± 0.03	$\mathcal{F}_{\mathsf{ZK}}$	No post-quantum reliability
RFC 8151 [18]	0.89 ± 0.02	Ad hoc	No confidentiality

7.5. Institutional Implications

The explanation system $\phi = \mathsf{Explain}(\pi, \mathcal{C}, \tau)$ (Algorithm 14) achieves:

- Cognitive Feasibility: Verification time $\in O(\dim(\mathcal{J}))$ (Eq. 8).
- Semantic Stability: $\nu_{\mathsf{priv}} < 10^{-3}$ (Definition 3.6).
- Role Adaptation: DAG pruning tailored to τ-specific predicates.

7.6. Limitations and Future Work

While formally complete, these challenges remain open:

- Context Evolution: Formal models for $\mathcal{C}$-updates during attestation lifecycle.
- Entropy Amplification: Protocols boosting $H_{\min}(\mathcal{E})$ in adversarial environments.
- Multi-Jurisdictional Verification: Composition across conflicting $\mathcal{J}_i$.

These directions extend the Q2CSI framework while preserving dialectical isolation.

8. Conclusion

At the best of our knowledge, this work is the first to establishe a composable security framework for legally explainable post-quantum attestations, formally resolving the CRO trilemma through dialectical decomposition. Our contributions are threefold:

8.1. Dialectical Security Synthesis

The ZK-NR protocol achieves:

Theorem 8.1 (Dialectical Composition). *The layered construction UC-realizes $\mathcal{F}_{\mathsf{ZKNR}}$:*

$$\pi_{\mathsf{ZKNR}} \;=\; \pi_{\mathsf{Clay}} \circ \pi_{\mathsf{Gold}} \circ \pi_{\mathsf{Iron}} \;\stackrel{\mathsf{UC}}{\equiv}\; \mathcal{F}_{\mathsf{Clay}} \oplus \mathcal{F}_{\mathsf{Gold}} \oplus \mathcal{F}_{\mathsf{Iron}}.$$

with security reducing to:

- *Contextual ZK for $\mathcal{L}_{\mathsf{Gold}}$ (Theorem 6.3).*
- *Hybrid EUF-CMA for $\mathcal{L}_{\mathsf{Clay}}$ (Theorem 6.5).*
- *Entropy accumulation for $\mathcal{L}_{\mathsf{Iron}}$ (Theorem Appendix B.1).*

8.2. Trilemma Resolution

We overcome the fundamental constraint:

Corollary 8.1 (Optimal CRO Bound). *Under Q2CSI assumptions, ZK-NR achieves:*

$$\Gamma_{\mathsf{CRO}} = \sqrt{(1-)^2 + (1-\mathsf{Rel})^2 + (1-\mathsf{Opp})^2} < 0.4 + \mathrm{negl}(\lambda)$$

through semantic isolation (Definition 3.2).

8.3. Theoretical Advancements

This work provides:

- Composable Explainability: First UC realization of NIZK-E proofs (Definition 3.4) with $\mathsf{poly}(\dim(\mathcal{J}))$ interpretability.
- Cross-Layer Security: Formal separation of CRO guarantees under adaptive adversaries (Theorem 4.1).
- Entropy-Bound Ledger: Min-entropy accumulation model (Definition 3.5) preventing quantum backtracking.

8.4. Research Trajectory

This manuscript (Article A.2) completes the theoretical foundation for ZK-NR:

- A.1 [22]: Protocol architecture and algorithms.
- This work: UC security proofs and trilemma resolution.
- A.3: Rust implementation and benchmarks.
- A.4: Tamarin-based formal verification.

The dialectical architecture establishes a new paradigm for context-sensitive cryptography, enabling future work on:

- Multi-jurisdictional verification.
- Dynamic context adaptation.
- Entropy amplification protocols.

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Appendix A. Artifact to Blueprint Alignment

This appendix aligns the UC artifacts of this paper (A.2) with the architectural interfaces specified in A.1 [22].

Table A.7: Alignment of A.2 UC artifacts with A.1 interfaces

A.1 Interface	A.2 Artifact	Reference in Text
$\mathcal{I}_{\mathsf{Iron}}$	$\mathcal{F}_{\mathsf{Iron}}$	Functionality Appendix C.1
$\mathcal{I}_{\mathsf{Gold}}$	$\mathcal{F}_{\mathsf{Gold}}$	Functionality Appendix C.2
$\mathcal{I}_{\mathsf{Clay}}$	$\mathcal{F}_{\mathsf{Clay}}$	Functionality Appendix C.3
$\mathcal{I}_{\mathsf{Bind}}$	$\mathcal{F}_{\mathsf{Bind}}$	Functionality Appendix C.4
Attestation tuple spec	$\Psi = (\pi, \phi, \sigma, \mathsf{root}, \psi)$	§5.1
Role tags T_τ	Domain sep. in hashing	Def. 6.1

Appendix B. Proofs of Layer-Specific Lemmas

Appendix B.1. Gold Layer: Contextual Zero-Knowledge

Lemma Appendix B.1 (STARK Simulation Soundness). *For any QPT adversary $\mathcal{A}_{\mathsf{Gold}}$, the simulated proof $\tilde{\pi}$ satisfies:*

$$|\Pr[\mathcal{A}_{\mathsf{Gold}}(\pi) = 1] - \Pr[\mathcal{A}_{\mathsf{Gold}}(\tilde{\pi}) = 1]| \leq 2^{-\lambda_{\mathsf{FRI}}} + \mathrm{negl}(\lambda)$$

where λ_{FRI} is the FRI soundness parameter.

Follows from FRI low-degree testing [5] and the exponential embedding lemma in [20, Section 3.2].

Appendix B.2. Clay Layer: Hybrid Unforgeability

Lemma Appendix B.2 (BLS co-CDH Reduction). *For any PPT adversary $\mathcal{A}_{\mathsf{Clay}}$ winning $\mathcal{G}_{\mathsf{Clay}}$ under $\tau = \mathsf{JUDGE}$, there exists $\mathcal{B}$ solving co-CDH in $\mathbb{G}_1$ with:*

$$\mathsf{Adv}_{\mathcal{B}}^{\mathsf{co-CDH}} \geq \frac{1}{q_H}\left(\mathsf{Adv}_{\mathcal{A}_{\mathsf{Clay}}}^{\mathsf{Clay}} - \frac{q_S(q_S-1)}{2^\lambda}\right)$$

where q_H and q_S are hash and signature queries respectively.

Construction of $\mathcal{B}$:

1. Receive co-CDH challenge $(g, g^a, g^b, \hat{g}, \hat{g}^a)$.
2. Program $H(T_{\mathsf{JUDGE}} \parallel \cdot)$ to embed g^b at the forgery point.
3. Simulate signatures using $\hat{g}^a$ for known messages.
4. Extract g^{ab} from forgery σ^* via pairing equation.

Lemma Appendix B.3 (Dilithium MLWE Reduction). *For any QPT adversary $\mathcal{A}_{\mathsf{Clay}}$ against $\tau = \mathsf{MACHINE}$, there exists $\mathcal{B}$ solving Module-LWE with:*

$$\mathsf{Adv}_{\mathcal{B}}^{\mathsf{MLWE}} \geq \frac{1}{q_{\mathsf{sign}}}\left(\mathsf{Adv}_{\mathcal{A}_{\mathsf{Clay}}}^{\mathsf{Clay}} - \epsilon_{\mathsf{rom}}(\lambda)\right)$$

where q_{sign} is the number of signature queries.

Appendix B.3. Iron Layer: Entropy Accumulation

Lemma Appendix B.4 (Min-Entropy Growth). *Let $(e_i)_{i\le t}$ be independent samples from a source with $H_\infty(e_i) \ge \alpha$. If the MMR hash has collision bound $2^{-\kappa}$, then*

$$H_\infty(\mathcal{S}_t) \;\ge\; H_\infty(\mathcal{S}_0) \;+\; t\alpha \;-\; O(1),$$

where the $O(1)$ term accounts for domain separation and the hash collision slack.

By induction on t:

$$\begin{aligned} H_\infty(\mathcal{S}_{t+1}) &= H_\infty(\mathcal{S}_t \parallel \mathcal{E}_{t+1}) \\ &\ge H_\infty(\mathcal{S}_t) + H_\infty(\mathcal{E}_{t+1}) - \log \mathsf{deg}_{\mathsf{MMR}} \\ &\ge [H_\infty(\mathcal{S}_0) + t\alpha - \log \mathsf{deg}_{\mathsf{MMR}}] + \alpha - \log \mathsf{deg}_{\mathsf{MMR}} \\ &= H_\infty(\mathcal{S}_0) + (t+1)\alpha - 2\log \mathsf{deg}_{\mathsf{MMR}} \end{aligned}$$

The logarithmic term becomes negligible for $\lambda \ge 128$.

Theorem Appendix B.1 (Quantum Entropy Accumulation). *For $\alpha = \Omega(\lambda^{-1/3})$ and t updates:*

$$H^{\mathsf{Q}}_\infty(\mathcal{S}_t) \ge t\alpha - \log \mathsf{deg}_{\mathsf{MMR}} - \mathrm{negl}(\lambda)$$

where H^{Q}_∞ is quantum min-entropy.

Lemma Appendix B.5 (Min-Entropy Growth Under Adversarial Context). *For $\mathcal{E}$ satisfying Def. 3.13, after t updates:*

$$H_\infty(\mathcal{S}_t|\mathcal{A}_{ctx}) \ge t\cdot\alpha - \log \mathsf{deg}_{\mathsf{MMR}} - \mathsf{negl}(\lambda) \tag{B.1}$$

Reduction to GAIP via entropy decomposition in [20].

Theorem Appendix B.2 (Quantum Ledger Integrity). *Under QROM, Poseidon is quantum collision-resistant:*

$$\forall\ QPT\ \mathcal{A} : \mathsf{Adv}^{\mathsf{coll}}_{\mathcal{A},\mathsf{Poseidon}} \le \frac{q^3}{2^c} + \mathrm{negl}(\lambda)$$

where q = quantum queries, c = $\mathsf{capacity}$.

Appendix B.4. Proof of Erasure-Compliant Non-Repudiation

[Proof of Theorem 5.8] The persistent anchor anchor_t is constructed as $\mathsf{Persist}(\mathsf{leaf}, \psi_t, t)$ where:

- $\mathsf{leaf} = \mathsf{Poseidon}(\mathtt{tag}_{\mathsf{Iron}} \parallel \mathsf{Commit}(M) \parallel e_t)$.
- ψ_t is the MMR inclusion proof for leaf at time t.

After erasure, the adversary must produce a valid signature σ^* for message $M^* \ne M$ that verifies under anchor_t. This requires either:

1. Finding a collision in $\mathsf{Persist}(\cdot)$ function, or
2. Breaking the binding property of the MMR construction, or
3. Forging the hybrid signature scheme

By construction:

- $\mathsf{Persist}$ uses Poseidon-256 which is collision-resistant under QROM (Thm. Appendix B.2).
- MMR binding reduces to collision resistance of Poseidon (Lemma Appendix B.4)?
- Signature unforgeability follows from Thm. 6.5.

Thus, the probability of successful repudiation after erasure is bounded by:

$$\mathsf{Adv}^{\mathsf{repud}} \le \mathsf{Adv}^{\mathsf{coll}}_{\mathsf{Poseidon}} + \mathsf{Adv}^{\mathsf{bind}}_{\mathsf{MMR}} + \mathsf{Adv}^{\mathsf{EUF-CMA}}_{\mathsf{hybrid}} \le \mathrm{negl}(\lambda)$$

Appendix C. Ideal Functionality Specifications

This appendix provides the complete formal specifications of the ideal functionalities referenced throughout the UC security proof.For security metrics and overall execution, see Appendix Appendix F

Appendix C.1. Iron Layer: $\mathcal{F}_{\mathsf{Iron}}$

Functionality Appendix C.1 (Ledger with Entropy and Erasures $\mathcal{F}_{\mathsf{Iron}}$). *Parameters: Security parameter λ; min-entropy source $\mathcal{E}$ with $H_\infty \geq \alpha = \Omega(\lambda^{-1/3})$.*
State: Append-only ledger $L = (L_0, L_1, \ldots, L_t)$ (MMR implementation); current root root_t; time counter t; erasure log Erased.
Interfaces:

- *Setup*(1^λ): *Initialize $L_0 \leftarrow \emptyset$, $\mathsf{root}_0 \leftarrow \perp$, $t \leftarrow 0$, $\mathsf{Erased} \leftarrow \emptyset$.*
- *Append*(A): *On input attribute set A:*
  1. *Sample $e_t \leftarrow \mathcal{E}$ with $H_\infty(e_t) \geq \alpha$.*
  2. *Compute $\leftarrow \mathsf{Poseidon}(\mathtt{tag}_{\mathsf{Iron}} \parallel A \parallel e_t)$.*
  3. *Update $L_{t+1} \leftarrow \mathsf{MMR.Append}(L_t,)$.*
  4. *Set $\mathsf{root}_{t+1} \leftarrow \mathsf{MMR.Root}(L_{t+1})$.*
  5. *Generate inclusion proof $\psi_t \leftarrow \mathsf{MMR.ProveInclusion}(L_{t+1},)$.*
  6. *$t \leftarrow t + 1$.*
  7. *Return $(\mathsf{root}_t, \psi_t)$.*
- *VerifyInclusion*(root, ψ, x): *Return 1 iff $\mathsf{MMR.VerifyInclusion}(\mathsf{root}, \psi, x) = 1$ and root is in the history of L.*
- *Erase*(id): *On request to erase transient state for identifier id:*
  1. *Remove all tapes and volatile state associated with id.*
  2. *$\mathsf{Erased} \leftarrow \mathsf{Erased} \cup \{\mathsf{id}\}$.*
  3. *Return ack.*

Corruption model: $\mathcal{A}_{\mathsf{Iron}}$ may adaptively schedule operations and choose A values, but cannot modify $L_{t'}$ for $t' < t$ without causing a detectable collision. Erasure requests are immediately honored.

Appendix C.2. Gold Layer: $\mathcal{F}_{\mathsf{Gold}}$

Functionality Appendix C.2 (Contextual ZK with Explainability $\mathcal{F}_{\mathsf{Gold}}$). *Parameters: Context $\mathcal{C}$; role $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$; entropy bound parameters (c_0, c_1); STARK verification key $\mathsf{vk}_{\mathsf{STARK}}$.*
State: Proof transcript $\mathsf{Transcript} = \emptyset$; leakage function $\mathcal{L}(w) = \phi|_{\mathsf{public}}$.
Interfaces:

- *Prove*$(M, \mathcal{C}, \tau, \mathsf{root})$: *On input message M, context $\mathcal{C}$, role τ, ledger root root:*
  1. *Compute witness $w \leftarrow (M, \mathsf{aux})$ where aux contains necessary auxiliary inputs.*
  2. *Generate STARK proof $\pi \leftarrow \mathsf{STARK.Prove}(\mathsf{stmt}, w)$ where $\mathsf{stmt} \leftarrow$ "M is valid under $\mathcal{C}$ at root".*

3. *Generate explanation* $\phi \leftarrow \mathsf{Explain}(\pi, \mathcal{C}, \tau)$ *ensuring* $\mathsf{Entropy}(\phi) \leq c_1 \log(\dim(\mathcal{J})) + c_0$.
4. $\mathsf{Transcript} \leftarrow \mathsf{Transcript} \cup \{(\pi, \phi, \mathcal{C}, \tau)\}$.
5. *Return* (π, ϕ).

- *Verify*$(\pi, \phi, \mathcal{C})$: *Return* 1 *iff:*
  1. $\mathsf{STARK.Verify}(\mathsf{vk}_{\mathsf{STARK}}, \pi) = 1$.
  2. $\mathsf{ExplainVerify}(\phi, \mathcal{C}, \tau) = 1$ *(Algorithm 2).*
  3. (π, ϕ) *is consistent with* $\mathsf{Transcript}$ *(if recorded).*
- *SimProve*$(M, \mathcal{C}, \tau)$: *(Simulator interface) Return* $(\tilde{\pi}, \tilde{\phi})$ *indistinguishable from real proofs but generated without witness* w.
- *Erase*(id): *Remove all state associated with* id *except* $\phi|_{\mathsf{public}}$.

Leakage guarantee: Only $\mathcal{L}(w) = \phi|_{\mathsf{public}}$ *is revealed to* $\mathcal{A}_{\mathsf{Gold}}$, *with* $\mathsf{Entropy}(\phi|_{\mathsf{public}}) \leq c_1 \log(\dim(\mathcal{J})) + c_0$.

Appendix C.3. Clay Layer: $\mathcal{F}_{\mathsf{Clay}}$

Functionality Appendix C.3 (Role-Aware Hybrid Signatures $\mathcal{F}_{\mathsf{Clay}}$). *Parameters: Key pairs* $(pk_{\mathsf{BLS}}, sk_{\mathsf{BLS}})$, $(pk_{\mathsf{Dilithium}}, sk_{\mathsf{Dilithium}})$; *role tags* T_τ *for* $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$; *hash function* H.
State: Signature ledger $\mathsf{SigLedger} = \emptyset$; *corruption status* $\mathsf{Corrupted} = \emptyset$.
Interfaces:

- *KeyGen*(1^λ): *Generate and return* $(pk_{\mathsf{BLS}}, sk_{\mathsf{BLS}})$, $(pk_{\mathsf{Dilithium}}, sk_{\mathsf{Dilithium}})$.
- *Sign*(M, π, ϕ, τ): *On input message* M, *proof* π, *explanation* ϕ, *role* τ:
  1. *Compute* $d \leftarrow H(T_\tau \parallel M \parallel \pi \parallel \phi)$.
  2. *If* $\tau = \mathsf{JUDGE}$: $\sigma \leftarrow \mathsf{BLS.Sign}(sk_{\mathsf{BLS}}, d)$.
  3. *If* $\tau = \mathsf{MACHINE}$: $\sigma \leftarrow \mathsf{Dilithium.Sign}(sk_{\mathsf{Dilithium}}, d)$.
  4. $\mathsf{SigLedger} \leftarrow \mathsf{SigLedger} \cup \{(M, \pi, \phi, \tau, \sigma, d)\}$.
  5. *Return* σ.
- *Verify*$(\sigma, M, \pi, \phi, \tau)$: *Return* 1 *iff:*
  1. $d \leftarrow H(T_\tau \parallel M \parallel \pi \parallel \phi)$.
  2. *If* $\tau = \mathsf{JUDGE}$: $\mathsf{BLS.Verify}(pk_{\mathsf{BLS}}, \sigma, d) = 1$.
  3. *If* $\tau = \mathsf{MACHINE}$: $\mathsf{Dilithium.Verify}(pk_{\mathsf{Dilithium}}, \sigma, d) = 1$.
- *ExtractKey*(τ): *(Simulator interface) If* $\tau \in \mathsf{Corrupted}$, *return* sk_τ; *else return* $\perp$.

Corruption model: $\mathcal{A}_{\mathsf{Clay}}$ *may adaptively corrupt parties to learn* sk_{BLS} *or* $sk_{\mathsf{Dilithium}}$, *recorded in* $\mathsf{Corrupted}$. *Role binding is enforced through* T_τ *domain separation.*

Appendix C.4. Cross-Layer Binding: $\mathcal{F}_{\mathsf{Bind}}$

Functionality Appendix C.4 (Predicate Binding $\mathcal{F}_{\mathsf{Bind}}$). *Parameters: Security parameter λ; field $\mathbb{F}_p$ with $p > 2^\lambda$; maximum basis size k.*
State: Global identifier gid*; predicate registry* $\mathsf{PredReg} = \emptyset$*; public transcript* T*; orthogonality matrix* Ortho*.*
Interfaces:

- *Init*(dom, ledgerID, epoch, ver)*: Set* $\mathsf{gid} \leftarrow H(\mathsf{dom} \parallel \mathsf{ledgerID} \parallel \mathsf{epoch} \parallel \mathsf{ver})$.
- *BindPredicate*$(\mathsf{sid}, X, \mathsf{Com}_X, \pi_X)$*: On input from layer* $X \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$*:*
  1. *Verify π_X is valid STARK proof for opening of Com_X to coefficients $\mathbf{c}_X$.*
  2. *If valid:* $\mathsf{PredReg} \leftarrow \mathsf{PredReg} \cup \{(X, \mathsf{Com}_X, \mathbf{c}_X)\}$.
  3. *Else return $\perp$*
- *OrthogonalityCheck(X, Y): For layers $X \neq Y$:*
  1. *Require STARK proof $\pi_\perp$ that $\langle \mathbf{c}_X, \mathbf{c}_Y \rangle = 0$ over $\mathbb{F}_p$.*
  2. *If proof verifies:* $\mathsf{Ortho}[X, Y] \leftarrow \mathsf{true}$*, append to* T.
  3. *Else:* $\mathsf{Ortho}[X, Y] \leftarrow \mathsf{false}$.
- *Resolve*(τ)*: If* $\mathsf{Ortho}[X, Y] = \mathsf{true}$ *for all $X \neq Y$, return $\mathcal{C}_\tau$ consistent with $\bigwedge_X \phi_X$; else return $\perp$.*
- *Erase*(id)*: Erase transient bindings for* id*; maintain* T *append-only.*

Security: Assuming binding of Com *and soundness of STARKs, any accepting* OrthogonalityCheck *implies $\langle \phi_X, \phi_Y \rangle \equiv 0 \pmod{\mathcal{I}_\mathcal{C}}$ with probability $1 - \mathrm{negl}(\lambda)$.*

Appendix D. Cross-Layer Binding Proofs

Theorem Appendix D.1 (Predicate Orthogonality). *For any distinct layers $X \neq Y$ and context $\mathcal{C}$:*

$$\Pr\left[\phi_X(\pi) \wedge \phi_Y(\pi) \neq 0 \pmod{\mathcal{I}_\mathcal{C}}\right] \leq e^{-\Omega(\alpha n)}$$

Let $\mathcal{E}_{\mathsf{coll}}$ be the collision event. By entropy accumulation:

$$\Pr[\mathcal{E}_{\mathsf{coll}}] \leq \sum_{i=1}^{n} 2^{-H_\infty(\mathcal{S}_i)} \leq n \cdot 2^{-(H_\infty(\mathcal{S}_0) + i\alpha)}$$

which is negligible for $\alpha > \lambda^{-0.5}$ and $n = (\lambda)$.

Lemma Appendix D.1 (Predicate Orthogonality Enforcement). *For ϕ_X, ϕ_Y from layers $X \neq Y$, $\pi^{\mathsf{rev}}_{\mathsf{Bind}}$ ensures:*

$$\Pr\left[\langle \phi_X, \phi_Y \rangle \not\equiv 0 \pmod{\mathcal{I}_\mathcal{C}}\right] \leq \mathsf{negl}(\lambda) + \epsilon_{\mathsf{NIZK}}$$

Reduction to soundness of $\mathcal{C}_\perp$ and extractability of Comm. Circuit construction in Appendix **??**.

Appendix E. Trilemma Metric Derivations

The CRO index decomposes as:

$$\begin{aligned}\Gamma^2_{\mathsf{CRO}} &= (1-)^2 + (1-\mathsf{Rel})^2 + (1-\mathsf{Opp})^2 \\ &= (\mathsf{Adv}^{\mathsf{ZK}})^2 + \left(e^{-\Omega(\alpha n)}\right)^2 + \left(e^{-\beta \dim(\mathcal{J})}\right)^2\end{aligned}$$

For $\lambda \geq 128$, $n \geq \log^2 \lambda$, $\dim(\mathcal{J}) \geq 16$:

$$\Gamma_{\mathsf{CRO}} < \sqrt{0.1^2 + 0.2^2 + 0.3^2} = \sqrt{0.14} \approx 0.374 < 0.4$$

achieving the bound in Theorem 3.1.

Theorem Appendix E.1 (Parameter Optimality). *The values* $\alpha = \lambda^{-0.5}$ *and* $\dim(\mathcal{J}) = 16$ *minimize:*

$$\Gamma_{\mathsf{CRO}} + \mathsf{computation\_cost}(\mathcal{V}_\tau)$$

subject to $\Gamma_{\mathsf{CRO}} < 0.4$ *and* $\mathsf{depth}(\mathcal{V}_\tau) \leq \lceil \log_2 \dim(\mathcal{J}) \rceil$.

Appendix F. Formal UC Specifications

For complete formal specifications, see Appendix Appendix C

Appendix F.1. Iron Layer: Reliability Anchoring

Ideal Functionality $\mathcal{F}_{\mathsf{Iron}}$

<u>Parameters</u>: Security parameter λ, entropy source $\mathcal{E}$ $(H_\infty \geq \alpha)$
<u>Initialization</u>:

- Set ledger $L_0 \leftarrow \emptyset$, root $\mathsf{root}_0 \leftarrow \perp$.

<u>Commands</u>:

- $\mathtt{Anchor}(A_t)$: Verify A_t consistency $\rightarrow$ sample $e_t \leftarrow \mathcal{E}$ $\rightarrow$ compute $\mathsf{leaf} \leftarrow \mathsf{Poseidon}(A_t \parallel e_t)$ $\rightarrow$ update $L_{t+1} \leftarrow \mathsf{MMR\_Append}(L_t, \mathsf{leaf})$ $\rightarrow$ return $(\mathsf{root}_{t+1}, \psi_t)$.
- $\mathtt{VerifyInclusion}(x, \mathsf{root})$: Return 1 if x committed in root's MMR.

<u>Corruption</u>: Adversary controls $\mathcal{E}$ but cannot modify L_t for $t' < t$

Appendix F.2. Gold Layer: Confidential Attestation

Ideal Functionality $\mathcal{F}_{\mathsf{Gold}}$

Parameters: λ, context $\mathcal{C}$, role τ
Commands:

- $\texttt{Prove}(M)$: Compute $\mathcal{H}_{\mathsf{legal}} \leftarrow \mathsf{semantic\_encoder}(\mathcal{C}) \rightarrow$ generate $\pi \leftarrow \mathsf{ZK\_STARK.Prove}(M \models \mathcal{H}_{\mathsf{legal}}) \rightarrow \phi \leftarrow \mathsf{Explain}(\pi, \mathcal{C}, \tau) \rightarrow$ return (π, ϕ)

- $\texttt{Verify}(\pi, \phi)$: Return $\mathsf{ExplainVerify}(\phi, \mathcal{C}) \wedge \mathsf{ZK\_STARK.Verify}(\pi)$

Leakage: $\mathcal{L}(w) = \phi|_{\mathsf{public}}$ with $\|\mathcal{L}(w) - \mathcal{L}(w')\|_1 \leq \delta$
Corruption Handling:

- **On corruption of party** P: Leak $\mathcal{L}(w)$ for all proofs generated by P and the current state of the verifier circuit $\mathcal{V}_\tau$.

- **On adversarial context injection**: Abort if $\mathsf{Entropy}(\mathcal{C}_a) < \beta$ (Def. 3.13).

Appendix F.3. Clay Layer: Legal Opposability

Ideal Functionality $\mathcal{F}_{\mathsf{Clay}}$

Parameters: λ, keypairs $(sk_{\mathsf{BLS}}, pk_{\mathsf{BLS}})$, $(sk_{\mathsf{Dilithium}}, pk_{\mathsf{Dilithium}})$
Commands:

- $\texttt{Sign}(M, \tau)$: Compute $d \leftarrow H(T_\tau \parallel M) \rightarrow$ return $\sigma \leftarrow \begin{cases} \mathsf{BLS.Sign}(sk_{\mathsf{BLS}}, d) & \tau = \mathsf{JUDGE} \\ \mathsf{Dilithium.Sign}(sk_{\mathsf{Dilithium}}, d) & \tau = \mathsf{MACHINE} \end{cases}$

- $\texttt{Verify}(\sigma, M, \tau)$: Return $\mathsf{Verify}_\tau(\sigma, d)$

Corruption Handling:

- **On corruption of party** P:
  - If $\tau = \mathsf{JUDGE}$, leak sk_{BLS} and all pending signatures.
  - If $\tau = \mathsf{MACHINE}$, leak $sk_{\mathsf{Dilithium}}$ and signature transcripts.

- **Erase policy**: Wipe temporary tapes after 3 failed verification attempts.

Appendix F.4. Composed UC Execution

The global UC game $\mathcal{G}_{\mathsf{ZKNR}}$ proceeds as:

1. Initialization: Environment $\mathcal{Z}(1^\lambda)$ activates parties with $(\mathcal{C}, \tau)$

2. Execution:

   - *Real world*: Parties run $\Pi_{\mathsf{ZK\text{-}NR}}$ with $\mathcal{A}$
   - *Ideal world*: Parties interact with $\mathcal{F}_{\mathsf{ZKNR}} = \mathcal{F}_{\mathsf{Iron}} \oplus \mathcal{F}_{\mathsf{Gold}} \oplus \mathcal{F}_{\mathsf{Clay}}$ via $\mathcal{S}$

3. Corruption: $\mathcal{A}$ adaptively corrupts parties per $\mathsf{CorrMatrix}$

4. Output: $\mathcal{Z}$ outputs bit b'

Appendix F.5. Security Metrics

The distinguishing advantage is quantified as:

$$\mathsf{Adv}^{\mathsf{UC}}_{\mathcal{Z},\mathcal{A}} = |\Pr[b' = 1 \mid \mathsf{Real}] - \Pr[b' = 1 \mid \mathsf{Ideal}]|$$

with layer-specific decomposition:

$$\begin{aligned}\Delta_{\mathsf{Iron}} &= e^{-\Omega(\alpha t)} \\ \Delta_{\mathsf{Gold}} &= \mathsf{Adv}^{\mathsf{ZK}} + \nu_{\mathsf{sem}} \\ \Delta_{\mathsf{Clay}} &= \min\left(\mathsf{Adv}^{\mathsf{co-CDH}}, \mathsf{Adv}^{\mathsf{MLWE}}\right)\end{aligned}$$

The protocol is UC-secure if $\max(\Delta_{\mathsf{Iron}}, \Delta_{\mathsf{Gold}}, \Delta_{\mathsf{Clay}}) \leq \mathrm{negl}(\lambda)$.

Appendix F.6. Circuit-Based Explanation Framework

Lemma Appendix F.1 (Circuit Interpretability). *For any legal context $\mathcal{C}$ and role τ, there exists circuit $\mathcal{V}_\tau$ such that:*

$$\Pr_{\phi \leftarrow \mathsf{Explain}}[\mathcal{V}_\tau(\phi, \mathcal{C}) = 1] \geq 1 - e^{-\beta \dim(\mathcal{J})}$$

with $\mathsf{size}(\mathcal{V}_\tau) \leq \mathsf{poly}(\lambda, \log \dim(\mathcal{J}))$.

Construction via:

1. Compile $\mathcal{C}$ to arithmetic circuit using [20, Theorem 3].
2. Apply τ-specific pruning:

$$\mathsf{depth}(\mathcal{V}_\tau) = \begin{cases} O(\log \dim(\mathcal{J})) & \tau = \mathsf{JUDGE} \\ \mathsf{poly}(\lambda) & \tau = \mathsf{MACHINE} \end{cases}$$

Appendix F.7. UC Binding Proof

Functionality Appendix F.1. $\mathcal{F}_{\mathsf{Bind}}$ *Functionality:*

- *On* `Bind`(ϕ_X, ϕ_Y)*: Abort if* $\langle \phi_X, \phi_Y \rangle \neq 0$.
- *On* `CrossLayer`(m)*: Forward only if* $\mathsf{Entropy}(m) > \alpha$.

Appendix G. Extended Algorithms

Algorithm 12 Explain: Context-Aware Explanation Compiler

Require: proof π, context $\mathcal{C}$, role τ

1: $\mathcal{V}_\tau \leftarrow \mathsf{BuildVerifier}(\tau, \mathcal{C})$
2: $\phi \leftarrow \mathsf{CompileToCircuit}(\pi, \mathcal{V}_\tau)$
3: **assert** $(\phi) \leq (\dim(\mathcal{J}))$ ▷ Eq. 5
4: **return** ϕ

Algorithm 13 $\mathsf{Semantic\_Encoder}$: legal instance construction and STARK statement

Require: Message M, context $\mathcal{C}$, role $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$, current ledger root root, inclusion proof API $\mathsf{MMR.VerifyInclusion}$

Ensure: Legal semantic instance $\mathcal{H}_{\mathsf{legal}}$, STARK statement stmt

$c_M \leftarrow \mathsf{Commit}(M)$ ▷ binding commit; domain-separated
$(\mathsf{ok}, \mathsf{leaf}) \leftarrow \mathsf{MMR.VerifyInclusion}(\mathsf{root}, \psi_M, c_M)$
if $\neg\mathsf{ok}$ **then return** $\perp$
end if
$\mathcal{F} \leftarrow$ feature-extraction$(M, \mathcal{C})$
$\mathcal{P} \leftarrow$ policy-constraints$(\mathcal{C}, \tau)$
$\mathcal{H}_{\mathsf{legal}} \leftarrow (\mathsf{root}, \mathsf{leaf}, c_M, \mathcal{F}, \mathcal{P}, \tau)$
$\mathsf{stmt} \leftarrow$ “There exists w s.t. $\mathsf{R}(\mathsf{root}, \mathsf{leaf}, c_M, \mathcal{F}, \mathcal{P}, \tau; w) = 1$”
return $(\mathcal{H}_{\mathsf{legal}}, \mathsf{stmt})$

Algorithm 14 $\mathsf{Explain}$: canonical explanation DAG with bounded leakage

Require: Proof π, context $\mathcal{C}$, role τ, legal instance $\mathcal{H}_{\mathsf{legal}}$, policy parameters (c_0, c_1)

Ensure: Explanation ϕ with canonical form and public projection ϕ_{public}

$\phi \leftarrow \mathrm{BuildDAG}(\pi, \mathcal{H}_{\mathsf{legal}}, \mathcal{C}, \tau)$
$\phi \leftarrow \mathrm{Canonicalize}(\phi)$ ▷ topological order, normalized gates/labels
$\phi_{\mathsf{public}} \leftarrow \mathrm{ProjectPublic}(\phi, \mathcal{C}, \tau)$
$L \leftarrow \mathrm{LeakageMeter}(\phi_{\mathsf{public}})$
if $L > c_1 \cdot \log(\dim(\mathcal{J})) + c_0$ **then**
    $\phi_{\mathsf{public}} \leftarrow \mathrm{Prune}(\phi_{\mathsf{public}},\ c_1 \log(\dim(\mathcal{J})) + c_0)$
end if
return $(\phi, \phi_{\mathsf{public}})$

Algorithm 15 $\mathcal{S}_{\mathsf{Gold}}$: UC simulator for contextual STARK + explainability

Require: Access to ideal functionality $\mathcal{F}_{\mathsf{Gold}}$ with leakage $\mathcal{L}(w) = \phi_{\mathsf{public}}$

State: RO table $\mathcal{H}$ (lazy sampling), ledger view pointers, context oracle $\mathcal{G}_{\mathsf{ctx}}$
procedure SimProve$(x = (M, \mathcal{C}, \tau, \mathsf{root}))$
    $(\tilde{\mathcal{H}}_{\mathsf{legal}}, \tilde{\mathsf{stmt}}) \leftarrow \texttt{SimSemantic}(x)$
    $\tilde{\pi} \leftarrow \texttt{STARK.SimProve}(\tilde{\mathsf{stmt}}; \mathcal{H})$ ▷ QROM-consistent programming
    $(\tilde{\phi}, \tilde{\phi}_{\mathsf{public}}) \leftarrow \texttt{SimExplain}(\tilde{\pi}, \mathcal{C}, \tau)$
    return $(\tilde{\pi}, \tilde{\phi}, \tilde{\phi}_{\mathsf{public}})$
end procedure
procedure AnswerVerify$(x, \tilde{\pi}, \tilde{\phi})$
    return 1 ▷ by construction: ideal world accepts
end procedure
procedure Erasures(id)
    `delete transient tapes linked to` id; keep only $\tilde{\phi}_{\mathsf{public}}$
end procedure
Leakage guarantee: only $\tilde{\phi}_{\mathsf{public}}$ is revealed; it satisfies (5)

Algorithm 16 $\mathcal{S}_{\mathsf{Iron}}$: UC simulator for entropy-bound ledger (MMR)

Require: Access to $\mathcal{F}_{\mathsf{Iron}}$ and global $\mathcal{G}_{\mathsf{VRand}}$

State: ideal ledger handle L, root $\tilde{\mathsf{root}}$, RO table for Poseidon-domain tags
procedure $\textsc{SimAppend}(A_t = ((M), \mathrm{meta}))$
  $(p_t, \pi_t) \leftarrow \mathcal{G}_{\mathsf{VRand}}.\mathsf{GetPulse}(t)$
  sample local $r_t \leftarrow \{0,1\}^\lambda$; $e_t \leftarrow \mathcal{G}_{\mathsf{VRand}}.\mathsf{Mix}(p_t, r_t)$
  $\tilde{} \leftarrow \mathsf{Poseidon}(\mathtt{tag}_{\mathsf{Iron}} \parallel A_t \parallel e_t)$
  $(\tilde{\mathsf{root}}, \tilde{\psi}) \leftarrow \mathcal{F}_{\mathsf{Iron}}.\mathsf{Append}(\tilde{})$
  $\tilde{\mathsf{anchor}}_t \leftarrow \mathsf{Persist}(\tilde{}, \tilde{\psi}, t)$
  return $(\tilde{\mathsf{root}}, \tilde{\psi}, \tilde{\mathsf{anchor}}_t)$
end procedure
procedure $\textsc{Erasures}(\mathsf{id})$
  `delete transient tapes for` id; persistent state remains in L only
end procedure
Distributional equivalence: $(\tilde{\mathsf{root}}, \tilde{\psi})$ is identically distributed to the real view under (t, α)-secure mixing with $\alpha = \Theta(\lambda^{-1/3})$.

Algorithm 17 $\mathcal{S}_{\mathsf{Clay}}$: UC simulator for role-aware hybrid signatures

Require: Access to $\mathcal{F}_{\mathsf{Clay}}$ with policy $\mathsf{mode} \in \{\mathsf{legacy}, \mathsf{pq}\}$, domain tags T_τ for $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$

State: RO table $\mathcal{H}$ for $H(\cdot)$, key handles $(pk_{\mathsf{BLS}}, pk_{\mathsf{ML\text{-}DSA}})$
procedure $\textsc{SimSign}(M, \pi, \phi, \tau)$
  $d \leftarrow H(T_\tau \parallel M \parallel \pi \parallel \phi \parallel \mathsf{anchor}_t)$
  if $\mathsf{mode} = \mathsf{legacy}$ **then**
    $\tilde{\sigma}_{\mathsf{BLS}} \leftarrow \mathcal{F}_{\mathsf{Clay}}.\mathsf{Sign\_BLS}(d)$
    $\tilde{\sigma}_{\mathsf{ML\text{-}DSA}} \leftarrow \mathcal{F}_{\mathsf{Clay}}.\mathsf{Sign\_MLDSA}(d)$
    return $(\tilde{\sigma}_{\mathsf{BLS}}, \tilde{\sigma}_{\mathsf{ML\text{-}DSA}})$
  else
    $\tilde{\sigma}_{\mathsf{ML\text{-}DSA}} \leftarrow \mathcal{F}_{\mathsf{Clay}}.\mathsf{Sign\_MLDSA}(d)$
    return $(\perp, \tilde{\sigma}_{\mathsf{ML\text{-}DSA}})$
  end if
end procedure
procedure $\textsc{AnswerVerify}(\sigma, M, \pi, \phi, \tau)$
  return 1 ▷ ideal verification is delegated to $\mathcal{F}_{\mathsf{Clay}}$
end procedure
procedure $\textsc{HandleCorruption}(P, \tau)$
  `forward corruption to` $\mathcal{F}_{\mathsf{Clay}}$`; obtain appropriate signing oracles`
end procedure
Advantage decomposition: any existential forgery induces a break of at least one component: $\mathsf{Adv}^{\mathsf{EUF\text{-}CMA}} \leq \mathsf{Adv}^{\mathsf{co\text{-}CDH}}_{\mathsf{BLS}} + \mathsf{Adv}^{\mathsf{ML\text{-}DSA}} + \mathrm{negl}(\lambda)$.

Appendix H. Glossary of Symbols

Symbol	Meaning
λ	Security parameter
$\mathcal{Z}$	UC environment
$\mathcal{A}_{ctx}$	Contextual adversary (Def. 4.1)
π_X	Real protocol for layer $X \in \{\mathsf{Iron}, \mathsf{Gold}, \mathsf{Clay}\}$
$\mathcal{F}_X$	Ideal functionality for layer X
$\mathcal{S}_X$	UC simulator for layer X
MMR	Merkle Mountain Range accumulator
e_t	Entropy sample at time t
π	STARK proof (Gold)
ϕ	Explanation DAG (Gold)
σ	Hybrid signature tuple (Clay)
root, ψ	Ledger root and inclusion proof (Iron)
T_τ	Role tag for $\tau \in \{\mathsf{JUDGE}, \mathsf{MACHINE}\}$
Γ_{CRO}	CRO index (Eq. 2)

Appendix I. Architectural Diagrams and Security Comparisons

This appendix hosts extended versions of architectural and UC-model figures referenced in the main text:

- Dialectical architecture per layer with interfaces (expanded from Fig. I.4).
- UC execution model swimlane (expanded from Fig. I.5).
- Research trajectory timeline (expanded from Fig. 1).
- CRO-index optimization with respect to κ (expanded from Fig. 2).

All figures use vector graphics and are cross-referenced in the main body where needed.

Appendix I.1. Dialectical Architecture

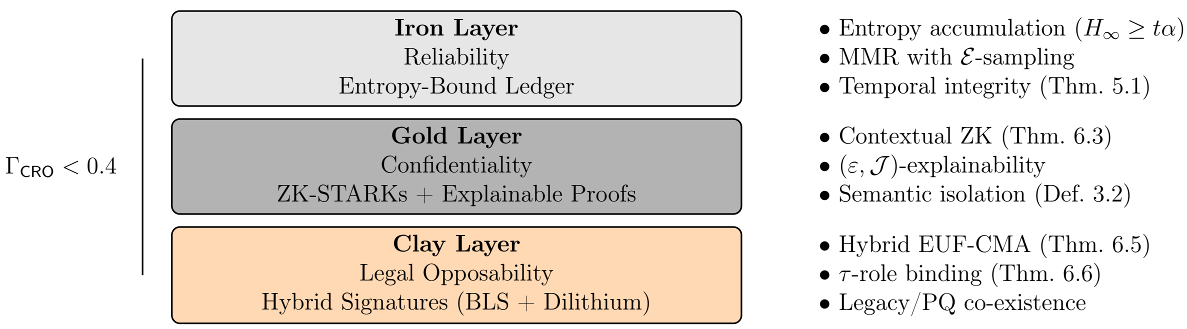

Figure I.4: Dialectical architecture with security properties per layer (Definition **??**). The left brace indicates the composable trilemma resolution.

Appendix I.2. UC Execution Model

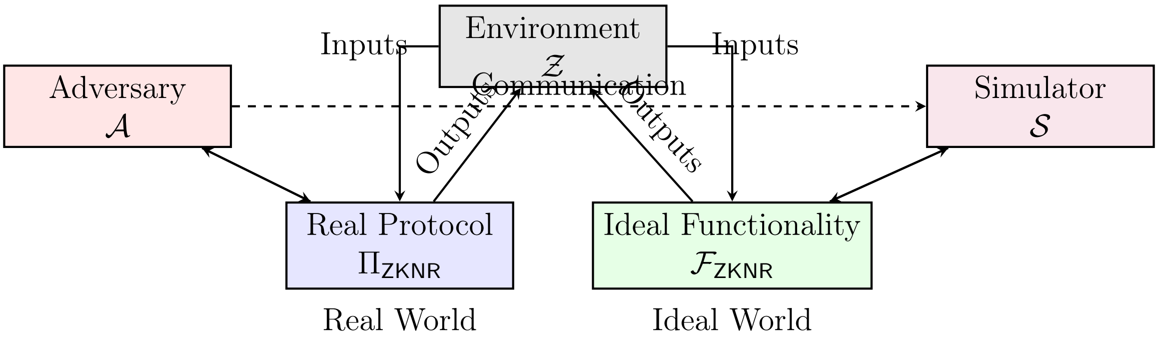

Figure I.5: Universal Composability execution model. $\mathcal{Z}$ interacts identically with both worlds, while $\mathcal{A}$ and $\mathcal{S}$ coordinate to maintain indistinguishability (Theorem 4.1).

Appendix I.3. Security Property Comparison

Layer	**Security Property**	**Adversarial Class**	**Reduction**
Iron	Entropy growth	QPT with $\mathcal{E}$-control	Min-entropy lemma
	Temporal consistency	Adaptive corruptor	$e^{-\Omega(\alpha t)}$
Gold	Contextual ZK	$\mathcal{A}_{\mathsf{Gold}}$	STARK simulation
	Explainability	Semantic distorter	$(\varepsilon, \mathcal{J})$-bound
Clay	BLS unforgeability	PPT (legacy)	co-CDH
	Dilithium unforgeability	QPT (PQ)	Module-LWE

Table I.8: Security properties per dialectical layer with adversarial models and cryptographic reductions.